Federated Malware Detection in Flying Ad Hoc Drone Networks Using Hybrid Convolutional Learning and Adaptive Optimization

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***Abstract*— The growing use of drones in edge computing, primarily within Flying Ad Hoc Networks (FANETs), has raised pressing security concerns, particularly in detecting malware within decentralized and privacy-sensitive environments. Traditional intrusion detection systems often fall short in these settings, struggling to meet the demands of limited computational power, non-IID data distributions, and the need to preserve local information. In response to these challenges, we propose F-ResDenseEffiNet, a novel federated deep learning framework that uniquely integrates ResNet, DenseNet, and EfficientNet. This architectural combination, not previously explored in FL-based malware detection, enables residual learning, feature reuse, and compound scaling, resulting in accurate and lightweight detection suitable for UAV platforms. To further improve convergence and adaptability in heterogeneous data settings, we introduce AGRO, an Adaptive Gorilla–Rabbit Optimization algorithm, which balances exploration and exploitation more effectively than standard optimizers such as Adam or RMSProp. Comprehensive experiments on four diverse intrusion detection datasets, DBMD24 (drone-malware), CICIDS2017, NSL-KDD, and KDDCup99—representing both traditional and UAV-relevant traffic patterns, showcase the model's strong performance, achieving an average detection accuracy of 97.9%, a threat identification rate of 99.7%, a decision confidence of 98.6%, and an overall classification quality of 98.7%. Additional evaluations, including ablation studies, statistical tests, fairness analysis, and architectural interpretability, further confirm the scalability, reliability, and suitability of the proposed framework for real-time drone-based anomaly detection.**

***Index Terms*— Federated learning, malware detection, drone networks, hybrid deep learning, edge security.**

I. Introduction

THE Internet is essential in contemporary life, particularly with the rise of Internet-enabled technologies like IoT [1]. IoT technology, which connects devices over the web, has improved living, working, and learning experiences. IoD stands for Internet of Drones, an extension of IoT [2]. Using IoD for UAV deployment is essential due to its "layered network control architecture" [3], [4]. Drones, UAVs, and FANET are synonymous in this work. Network security research is crucial due to the prevalence of the Internet and advanced communication technologies. Data and network security need IDSs and firewalls. NIDS monitors all network traffic for malicious activity. In the last decade, communications and network technologies have rapidly expanded network size and data sharing, making new threats more challenging to detect. A compromise at a single node might impact an organization's data. The US Department of Transportation assessed the viability of civil GPS spoofing, including drone interference [6]. Signatures, anomalies, and specifications can trigger IDSs. Anomaly-based methods involve knowledge, machine learning, and statistics. Detecting abnormal intrusions involves comparing traffic data to a model of ordinary operation [7], [8]. Machine-learning anomaly detection systems enhance performance by refining models based on observed patterns. Signature-driven identification compares current data to threat trends and eliminates false alarms, but requires regular signature updates [9].

Machine learning algorithms dominate this article. They work better than non-machine learning ones because they learn and teach. Drone networks require reliable intrusion detection. Researchers suggest using deep learning (DL) and machine learning (ML) to extract meaningful data from large datasets [10]. Network security uses ML and DL approaches to learn from network data and distinguish between normal and unusual activities. Unlike feature engineering-based machine learning, deep learning can immediately learn complex features from raw data. Over time, researchers have created ML and DL-based NIDS risk identification algorithms. NIDS must cope with increased security concerns and network traffic. According to [11] and [12], the agent monitors the network, while the analysis engine and response module detect attacks and report findings. Despite improvements, standard NIDSs are inadequate for IoT's complex network layers due to the critical applications of drone operations and security considerations. The addition of UAVs introduces complexity, necessitating enhanced intrusion detection.

This study presents a federated deep learning framework designed explicitly for malware detection within the Internet of Drones (IoD) environment. At its core is a hybrid model called F-ResDenseEffiNet, which integrates the strengths of ResNet, DenseNet, and EfficientNet to achieve high detection accuracy while staying within the computational and energy constraints of UAV platforms. Each drone is equipped with a lightweight version of the model that operates independently to identify

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local threats in real time. Meanwhile, a central base station aggregates insights across the network to coordinate a broader, swarm-wide response. What sets this approach apart from traditional IoT-based solutions is its alignment with the operational realities of drone systems—limited bandwidth, high mobility, and strict privacy requirements. By leveraging federated learning, the model avoids transmitting raw data, keeping sensitive information on each drone while still enabling collaborative intelligence. To further address the challenges of uneven data distribution and sporadic connectivity standards in Flying Ad Hoc Networks (FANETs), the system incorporates AGRO (Adaptive Gorilla–Rabbit Optimization). This dual-phase optimization method ensures stable and efficient learning. Evaluations across four benchmark datasets confirm that the proposed method delivers strong performance and real-world viability for secure, real-time anomaly detection in drone networks.

Research Aim: The primary focus of this research is to design a privacy-preserving federated deep learning model capable of detecting malware efficiently in real-time UAV network environments. The proposed solution is designed to respect privacy constraints while maintaining high detection accuracy, efficient resource utilization, and scalability across distributed UAV systems.

Research Questions:

- **RQ1:** What design considerations are necessary to optimize deep hybrid models for malware classification on drones with limited processing capabilities?
- **RQ2:** How can federated learning be leveraged to enable secure and collaborative model training in drone environments, particularly under non-IID data conditions?
- **RQ3:** Which optimization strategies are most effective for accelerating convergence and improving accuracy in decentralized UAV systems facing unreliable communication and dynamic topologies?

Significance: These research questions address core challenges in securing the Internet of Drones (IoD). As drones increasingly support critical missions in areas such as logistics, monitoring, and defense, there is a pressing need for lightweight, adaptable, and privacy-aware cybersecurity frameworks. Meeting these demands will help ensure resilient operations and safeguard sensitive data in complex, real-world deployment scenarios.

Key Contributions: The principal contributions of this research are outlined below:

1) In this research, we introduce a federated learning framework that can detect anomalies in FANETs, flying ad hoc networks that utilise drones in real-time. This system features asynchronous training, adherence to stringent data privacy standards, and optimisation for deployment on edge-level platforms. The proposed method ensures improved scalability and resilience in decentralised UAV settings, in contrast to centralised FL-CNN-based systems.
2) A hybrid deep learning model, termed F-ResDenseEffiNet, is introduced by integrating the residual feature propagation capabilities of ResNet, the dense connectivity principles of DenseNet, and the compound scaling efficiency of EfficientNet. Unlike prior approaches such as [30] that rely on graph neural networks in combination with convolutional models, this architecture is meticulously optimized to achieve high inference accuracy while maintaining compatibility with the constrained computational profile of drone hardware.
3) A novel optimization method, Adaptive Gorilla–Rabbit Optimization (AGRO), is proposed to improve convergence dynamics under non-IID data settings. This algorithm uniquely merges exploratory behavior inspired by gorilla search heuristics with the fast convergence traits observed in rabbit-based dynamics. It is especially well-suited for federated learning contexts characterized by unstable network connectivity and heterogeneous client data, as frequently encountered in aerial communication networks.
4) The efficacy and generalizability of the framework are demonstrated through extensive validation across four benchmark intrusion detection datasets: DBMD24, CICIDS2017, NSL-KDD, and KDDCup99. These datasets represent a broad spectrum of real-world malware behaviors and attack scenarios, enabling comprehensive evaluation under diverse conditions.
5) A thorough experimental analysis is conducted, encompassing ablation studies, statistical hypothesis testing, fairness assessments, and hyperparameter sensitivity evaluations. The proposed system consistently delivers an average detection accuracy of 98.9 percent, underscoring its robustness, adaptability, and practical utility for real-time malware detection in UAV-based surveillance and monitoring operations.

The structure of this article is organized to guide the reader through each stage of the research. The second section looks at the works that are most closely related. In Section III, the suggested organization is summed up. Section IV outlines the experiments and assessment criteria employed, and Section V analyzes and comments on the simulation results. Section 6 concludes the study and provides ideas for future projects.

II. Related Work

Recent breakthroughs have significantly enhanced malware detection capabilities within IoT and drone-based networks, leveraging deep learning and edge intelligence. Traditional methods, which often depend on centralized architectures, struggle to address critical concerns such as data privacy, communication overhead, and deployment scalability. In response to these limitations, A decentralized solution that allows remote nodes to learn together without exchanging raw data is Federated Learning (FL) [13]. This section reviews both federated and non-federated approaches, analyzing each study in terms of its objectives, core methodology, performance achievements, and constraints, with a focus on their applicability to secure, privacy-aware, and resource-constrained drone environments.

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To detect intrusions in IoT networks, [14] developed a deep learning model that incorporated CNNs and LSTM networks. The method's accuracy rate of 96.2% and F1-score of 0.965 on the IoT-23 dataset were promising. Still, it was only suitable for centralised processing, which rendered it ineffective for distributed applications like drone swarms, where privacy is essential. A strategy presented in [15] utilized time-window embeddings in conjunction with a feedforward neural network and a transformer encoder to identify anomalies in smart homes enabled by the Internet of Things. On the Aposemat IoT-23 dataset, the system achieved an F1-score of 0.937. Despite its strong results, the model was developed for centralized environments and lacked flexibility for deployment in decentralized aerial systems. In [16], a lightweight ensemble-based model was proposed for botnet detection across multiple IoT datasets, including IoT-23. It achieved a binary classification accuracy of 97.9%. While computationally efficient, the method does not support federated learning and has not been validated in dynamic drone networks with real-time constraints. A GRU-based botnet detection model presented in [17] demonstrated exceptional accuracy of 99.89% on IoT-23. Although effective, the model's reliance on centralized learning hinders its suitability for decentralized drone-based ecosystems where privacy and on-device inference are critical.

A notable federated learning-based approach was proposed in [18], utilizing depthwise separable CNNs to detect traffic anomalies in IoT networks. The model demonstrated strong performance, achieving 98.52% accuracy for binary classification and 97.73% for multiclass scenarios on the IoT-23 dataset. While this work highlights the privacy benefits of FL, it does not address key deployment challenges often encountered in drone networks, such as vulnerability to adversarial attacks or unstable connectivity. As outlined in the foundational study by [13], federated learning is particularly well-suited for decentralized and privacy-sensitive environments, such as UAV swarms, where sharing raw data is restricted due to bandwidth constraints and security concerns. Their insights into communication efficiency, heterogeneity handling, and privacy-preserving strategies directly support our decision to adopt FL as the backbone of our malware detection framework in Flying Ad Hoc Networks (FANETs).

In [19], conventional classifiers like Random Forest, Multi-Layer Perceptron, and Gradient Boosting were applied to the IoT-23 dataset, with Random Forest yielding the best result at 98.6% accuracy. The study, however, remained centralized and did not account for edge-device constraints or the privacy-preserving benefits of FL. A temporal model named ADEPT was introduced in [20] for phase-wise attack detection, achieving 99% accuracy and an F1-score of 0.99 using NSS and IoT-23 datasets. Despite its effectiveness, the absence of federated capabilities limits its relevance for drone-based distributed learning environments. A federated CNN-LSTM architecture was implemented in [21] and [22] to detect wormhole attacks across distributed nodes using the IoT-23 dataset. The model preserved privacy and achieved 96% accuracy, although its vulnerability to adversarial attacks and poisoned updates was not addressed.

The study in [23] and [24] compared centralized and federated versions of SVM and LSTM models on the VirusTotal dataset. The federated LSTM variant achieved the highest accuracy at 91.67%, striking a balance between privacy and performance. However, it lacked mechanisms to handle noisy or non-IID client data, an essential factor in drone applications. In [25], a federated framework utilizing both supervised and unsupervised models was applied to the N-BaIoT dataset for malware detection. It maintained strong accuracy, exceeding 97%, while preserving user privacy. Yet, the study did not consider heterogeneous data distributions across clients—a realistic challenge in drone swarms. A comparative evaluation in [26] used over 87,000 Android applications from the Opera Mobile Store to test centralized and federated SVM models. Although the federated model had slightly lower accuracy (93.45%) compared to centralized (94.05%), it outperformed in precision and recall. However, it lacked hybrid deep learning components that could further enhance detection capabilities. In [27], a federated malware detection model based on the "Less is More" framework was evaluated on the MaMaDroid dataset, which included 200 users. The model achieved an F1-score of 95%. While privacy-aware, the method required significant preprocessing and lacked deeper neural architectures for feature extraction. The FedHGCDroid framework in [28] introduced a federated approach that integrates Convolutional Neural Networks (CNNs) with Graph Neural Networks (GNNs) for malware detection. Evaluated on the Androzoo dataset, it achieved a notable accuracy of 91.3% and demonstrated effective adaptability across distributed clients. Deploying it on drone platforms with limited resources may be challenging due to its relatively high processing requirements, despite its performance.

Incorporating generative adversarial networks for Android malware detection via federated learning was part of the Fed-IIoT architecture in [29]. With a performance reaching 93.24% accuracy on the Genome dataset, the system showed promise; however, training instability due to GANs and the presence of non-IID data pose implementation challenges. The authors of [30] introduced FedMalDE, a federated learning framework that use a Subgraph Aggregated Capsule Network (SACN), in an effort to enhance malware detection. The model attained a remarkable F1-score of 97.64% when evaluated on datasets like DREBIN and AndroZoo. However, its reliance on graph-based structures adds computational complexity, which could hinder real-time implementation in drone systems where low latency and resource efficiency are critical. In [28], a comparison of tree-based classifiers such as Random Forest, CatBoost, and LightGBM was conducted using the CIC-MalMem-2022 dataset. Random Forest achieved near-perfect accuracy, but the centralized framework lacks support for federated inference or secure edge deployment. A hybrid ML-DL approach was proposed in [31], combining SMOTE and XGBoost for intrusion detection across CIC-MalMem-2022 and KDDCUP-99. Achieving 100% accuracy, the system performed well but lacked any distributed training or data protection capabilities.

An evaluation of clustering and classification techniques, including AdaBoost, K-means, and DBSCAN, was performed

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TABLE I
SUMMARY OF RECENT LITERATURE CATEGORIZED BY MODEL AND DEPLOYMENT STRATEGY

Ref	Problem Analyzed	Method Used	Achievement	Limitations
1. Centralized Deep and Machine Learning Models				
[14]	Intrusion detection in IoT environments using temporal models	CNN + LSTM	96.2% accuracy, 0.965 F1-score (IoT-23)	Centralized; lacks privacy and scalability
[15]	Anomaly detection in smart home IoT data streams	FFNN + Transformer	0.937 F1-score (IoT-23)	Not suited for distributed environments
[16]	Lightweight botnet detection in IoT	Ensemble learning	97.9% binary accuracy	No FL support, untested on drone data
[17]	Sequence modeling for botnets	GRU	99.89% accuracy (IoT-23)	Centralized; no privacy mechanisms
[19]	Attack classification via traditional ML	RF, MLP, GB	98.6% accuracy (IoT-23)	Not designed for edge or FL scenarios
[20]	Multi-phase attack detection	ADEPT framework	99% accuracy, 0.99 F1 (IoT-23 + NSS)	No federated or privacy consideration
[31]	Hybrid ML-DL for intrusion detection	SMOTE + XGBoost + DL	100% accuracy (CIC-MalMem-2022)	No privacy-aware or distributed execution
[32]	Unsupervised + ensemble classification	AdaBoost, DBSCAN, etc.	99.99% accuracy (CIC-MalMem-2022)	No multi-client or FL support
[33]	Memory-resident malware detection	Extended CNN	99% detection, 83% classification (CIC-MalMem-2022)	Centralized; not drone-compatible
2. Federated Learning with Classical Models				
[23]	Malware classification comparison	FL vs Central SVM + LSTM	91.67% (FL LSTM, VirusTotal)	No personalization; data heterogeneity issues
[25]	FL for IoT malware detection	Supervised + unsupervised FL	¿93% accuracy (N-BaIoT)	Client variability not explored
[26]	Android malware across apps	FL SVM vs centralized	93.45% accuracy (Opera APKs)	No hybrid model use; slight drop in FL
[27]	Privacy-aware malware classification	FL with LiM framework	95% F1 (MaMaDroid)	Heavy preprocessing; lacks deep features
3. Federated Learning with Deep or Hybrid Models				
[18]	FL anomaly detection in IoT traffic	Fed. CNN + Depthwise CNN	94.52% (binary), 97.73% (multiclass)	Lacks drone adversarial resilience
[21]	Wormhole attack detection	FL CNN + LSTM	96% accuracy (IoT-23)	Robustness to attacks not tested
[28]	Android malware with hybrid GNNs	FedHGCDroid (CNN + GNN)	91.3% accuracy (Androzoo)	High computation load for drones
[29]	Android malware with GANs	FedGAN-BM / FedGAN-BK (Fed-IIoT)	93.24% accuracy (Genome)	GAN instability in non-IID data
[30]	Semi-supervised malware detection	FedMalDE with SACN	95.64% F1 (4 datasets)	High complexity for swarm deployment
4. Feature Optimization and Selection				
[34]	Optimization for malware features	PSO, BBA, MA + RF/SVM	89% accuracy (CIC-MalMem-2022)	Not integrated with FL or secure edge training

in [32], with AdaBoost achieving 99.99% accuracy on CIC-MalMem-2022. Though effective, these models were not adapted for federated training or collaborative learning scenarios. The extended CNN model introduced in [33] focused on detecting memory-resident malware, reporting 99% detection accuracy and 83% accuracy in malware type classification on CIC-MalMem-2022. Despite a solid performance, its centralized structure prevents practical application in privacy-sensitive drone operations. Finally, the study in [34] employed metaheuristic algorithms, including Particle Swarm Optimization (PSO), Binary Bat Algorithm (BBA), and Mayfly Algorithm (MA), for feature selection in malware detection. Combined with classifiers such as SVM and Random Forest, the framework achieved approximately 99% accuracy on CIC-MalMem-2022. However, its lack of federated capabilities restricts use in distributed or privacy-critical systems.

The need to include privacy-preserving methods in both federated and decentralised intrusion detection systems has been highlighted in recent research. For example, [35] presents research that utilises the Laplace mechanism and statistical metrics, such as information value and weight of evidence, to develop a differential privacy strategy for deep neural networks. A significant trade-off between privacy and usefulness is maintained. At the same time, this strategy successfully preserves both numerical and categorical sensitive data. Despite its effectiveness in privacy preservation, the model's lack of emphasis on real-time processing and resource optimization limits its suitability for UAV-based deployment. Similarly, the research in [36] proposes an advanced TB-MFCC-based multifuse feature extraction framework aimed at classifying emergency vehicle sounds. By combining signal augmentation techniques with a Multi-Stacked CNN and an Attention-enhanced BiLSTM network, the model achieves a high classification accuracy of 98.66%. Although not directly related to malware detection, the study offers valuable insights into handling data imbalance and extracting rich temporal-spatial features—strategies that can be adapted to drone-based threat identification systems. In another contribution, [37]

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presents a hybrid privacy-preserving model that merges statistical transformations with a privacy-chain-based homomorphic encryption scheme. Applied across diverse datasets, including lung cancer and bank marketing data, the model demonstrates high classification accuracy while protecting user-sensitive information. However, the encryption process incurs significant computational overhead, which can pose practical limitations for deployment in low-power, real-time UAV environments.

Another study focused on strengthening the security of drone and Internet of Drones (IoD) environments through advanced intrusion and malware detection techniques. To enhance anomaly detection in drone-to-drone and drone-to-base station communications, the Cross-Layer Convolutional Attention Network (CLCAN) was proposed in the research by [38]. This network utilises contextual attention, multi-scale convolutional layers, and dynamic feature fusion. Their approach reported detection accuracy below 96% on real drone communication datasets. Work presented in [39] explored artificial intelligence–based methods for malware detection in drone communication, proposing a hybrid model with feature selection (Hybrid with FS). Although it improved performance, accuracy remained around 80%, underscoring the need for more robust solutions. A ConvLSTM-based approach was examined in [40], where the method was validated across benchmark datasets such as KDDCup99, NSL-KDD, CICIDS2017, and WSN-DS, achieving performance slightly below 96% across different attack categories. The identification of Denial of Service (DoS) attacks in Internet of Things (IoT) settings was examined by [41], who also detailed the advantages and disadvantages of UAV intrusion detection systems that utilise machine learning. Research from these studies indicates that drone security solutions powered by AI have considerable potential. Still, they also have specific common problems, such as reliance on datasets and limited attack coverage. Building on these insights, our work introduces a federated hybrid deep learning framework that not only strengthens malware detection in drone networks but also demonstrates consistent generalization across diverse benchmark datasets (NSL-KDD, KDDCup99, CICIDS2017).

Although earlier studies have made meaningful contributions to the development of federated learning frameworks and malware detection methods, many still overlook the unique combination of challenges that arise in practical drone-based environments. Many existing models rely on single-path architectures, such as CNNs or LSTMs, and often do not leverage hybrid designs or adaptive tuning techniques. Furthermore, only a few approaches have been tested under drone-specific conditions such as limited connectivity, exposure to adversarial threats, or real-time processing constraints on lightweight hardware. To address these limitations, the proposed F-ResDenseEffiNet model combines the strengths of ResNet, DenseNet, and EfficientNet to create a compact yet powerful architecture tailored for UAV platforms. The AGRO algorithm complements this—a hybrid Gorilla–Rabbit optimization strategy—that enhances convergence and generalization, particularly under the non-IID data and intermittent communication settings typical in federated UAV networks. An aggregated summary of the existing literature is provided in Table I, while a detailed comparative analysis focusing on key differentiators such as federated capability, drone compatibility, optimization mechanisms, and resilience to non-IID conditions is presented in Table II.

III. Proposed System Model

The proposed system is structured around a well-defined design cycle comprising six interconnected stages. The process begins with the collection of data from drone-mounted sensors and network traffic sources, capturing key operational and environmental variables, including IP addresses, packet contents, GPS coordinates, and surrounding conditions. Once gathered, the raw data is preprocessed to ensure quality—this involves cleaning, normalization, and addressing any missing values, making the data reliable and ready for analysis.

In the third stage, feature engineering is applied to extract insightful attributes, such as traffic volume, packet flow length, and variations in packet size—features that are crucial for identifying malware patterns. Statistical and machine learning techniques are employed in a hybrid feature selection strategy, which is utilised in the fourth phase to identify the most crucial features. The data will be simplified, and the model's performance will be improved.

The next step is to feed the ResDenseEffiNet classifier the cleaned-up dataset. This solution enables multiple drones to train together in a federated learning context without exchanging raw data, thereby respecting user privacy and reducing transmission costs. Step six involves introducing the Adaptive Gorilla-Rabbit Optimisation (AGRO) approach to finish optimising the model's hyperparameters. AGRO excels in providing constant and fast convergence, even in the most challenging circumstances, such as FANETs' notoriously non-iid data and intermittent network connections.

The hybrid ResDenseEffiNet model serves as the core intelligence of the system, offering a carefully crafted balance between architectural depth, representational power, and computational efficiency. By integrating ResNet's residual learning, DenseNet's feature reuse, and EfficientNet's compound scaling, the model achieves high detection accuracy while minimizing the computational demands on onboard UAV hardware. Empirical evaluations indicate that the model requires approximately 305 MFLOPs per inference and occupies 14.2 MB of memory, well within the operational limits of embedded systems such as the NVIDIA Jetson Nano and Qualcomm Flight RB5. When compared to established lightweight models like MobileNetV2 (238 MFLOPs, 13.6 MB) and SqueezeNet (833 MFLOPs, 4.8 MB), ResDenseEffiNet achieves a consistent 6–9% improvement in accuracy, with only a marginal increase in inference latency (62 ms versus 49 ms). This performance-to-resource trade-off makes the model a viable and deployable solution for real-time malware detection in UAV environments.

Importantly, only the trained ResDenseEffiNet model is deployed on drones for inference tasks, while the federated training process is executed asynchronously across distributed edge nodes. This separation ensures that drones are not burdened with training overhead during active operation, thereby

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TABLE II
COMPARATIVE ANALYSIS OF EXISTING METHODS AND PROPOSED FRAMEWORK

Method	Federated Learning Support	UAV Deployability	Optimization Strategy	Non-IID Robustness
CNN + LSTM [14]	✗	✗	✗	✗
Transformer + FFNN [15]	✗	✗	✗	✗
Ensemble Lightweight Model [16]	✗	✓	✗	✗
GRU-Based Botnet Detection [17]	✗	✓	✗	✗
Fed-CNN with Depthwise Layers [18]	✓	✓	✗	✗
ADEPT Framework [20]	✗	✓	✗	✗
FL-CNN + LSTM [21]	✓	✓	✗	✗
FedHGCDroid (CNN + GNN) [28]	✓	✗	✗	✓
FedGAN-BM [29]	✓	✗	✓	✗
FedMalDE (SACN) [30]	✓	✗	✗	✓
Metaheuristic + RF/SVM [34]	✗	✗	✓	✗
F-ResDenseEffiNet (Proposed)	✓	✓	✓(AGRO)	✓

preserving responsiveness and maintaining a practical balance between processing cost and real-time detection performance.

To assess its robustness, the model was validated across four well-known intrusion detection datasets—DBMD24 [42], CICIDS2017 [44], NSL-KDD, and KDDCup 99 [43]—covering a wide range of cyberattack scenarios and network behaviors. As illustrated in Figure 1, the proposed architecture supports distributed analysis, where individual drones detect local anomalies. At the same time, the base station aggregates updates to maintain global security awareness across the network.

A. Stage 1: Dataset Description

This study's dataset is built to help detect malware by analyzing ambient variables, network traffic, data from drone GPS and sensors, and metrics for malware identification. Drone operations and possible cyber risks are both modeled in this dataset. The "Drone-based Malware Detection (DBMD)" dataset [42], available on Kaggle at [29], was used to acquire the data. The dataset comprises 54 attributes and 200,000 records, derived from four distinct categories. Some of the attributes are shown in Table III. The dataset used in this study is publicly available. It is intended solely to advance research in secure drone communication and malware detection. Nevertheless, as with many cybersecurity datasets, there is a potential for misuse if not properly applied. Therefore, its use should remain strictly within the bounds of ethics and the law.

1) Derived Features: We created various additional characteristics from the current ones to strengthen the dataset and raise the efficiency of the malware detection algorithms. These generated characteristics find further trends and connections in the data. The derived features are described in Table IV.

B. Stage 2: Data Preprocessing

The first and foremost stage in preparing data for training machine learning algorithms is data preparation. Preprocessing can ensure the integrity and consistency of raw input fields and prepare them for better model performance(iterative & more accurate results). The drone-based malware detection dataset was preprocessed with the following steps [45]:

TABLE III
SUBSET OF KEY FEATURES FROM EACH CATEGORY

Category	Feature Name	Description
Network Traffic	timestamp	Timestamps for each network activity record
	source_ip	IPv4 addresses of the source devices
	destination_ip	IPv4 addresses of the destination devices
	protocol	Network protocols (TCP, UDP, ICMP)
	packet_length	Length of network packets (bytes)
	traffic_volume	Volume of traffic in bytes
	...	...
Drone GPS and Sensor	latitude	Location data of the UAV
	longitude	drone's Longitude coordinates
	altitude	altitude of the UAV (meters)
	speed	Speed of the drone (m/s)
	battery_level	Battery level of the drone (%)
	...	...
Malware Detection	anomaly_score	Anomaly score (0 to 1)
	suspicious_ip_count	number of suspicious IPs detected
	malicious payload indicator	Indicator for malicious payloads (0 or 1)
	reputation_score	Reputation score (0 to 100)
	attack_type	Type of malware attack
	...	...
Environmental Data	location_type	Type of location (urban or rural)
	nearby_devices	Number of nearby devices
	signal_interference	Signal interference level
	time_of_day	Time of day
	weather_conditions	Weather conditions (sunny, rainy, cloudy, stormy)
	...	...

Handling Missing Values: The first step in maintaining a clean dataset is to replace any missing values. When numerical characteristics had missing values, the mean of that feature was used to fill them in, ensuring the data remained normally distributed. The distribution of categorical characteristics was maintained by imputing missing values using the mode.

Normalization and Standardization The numerical characteristics were standardized and normalized to guarantee their equal scale. Algorithms that depend on distance measurements rely on this stage. Min-max scaling normalized features to [0, 1] and StandardScaler to zero-average and zero-deviation. The process of standardization and normalization can be stated as follows:

$$x' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \tag{1}$$

$$x'' = \frac{X - \mu}{\sigma} \tag{2}$$

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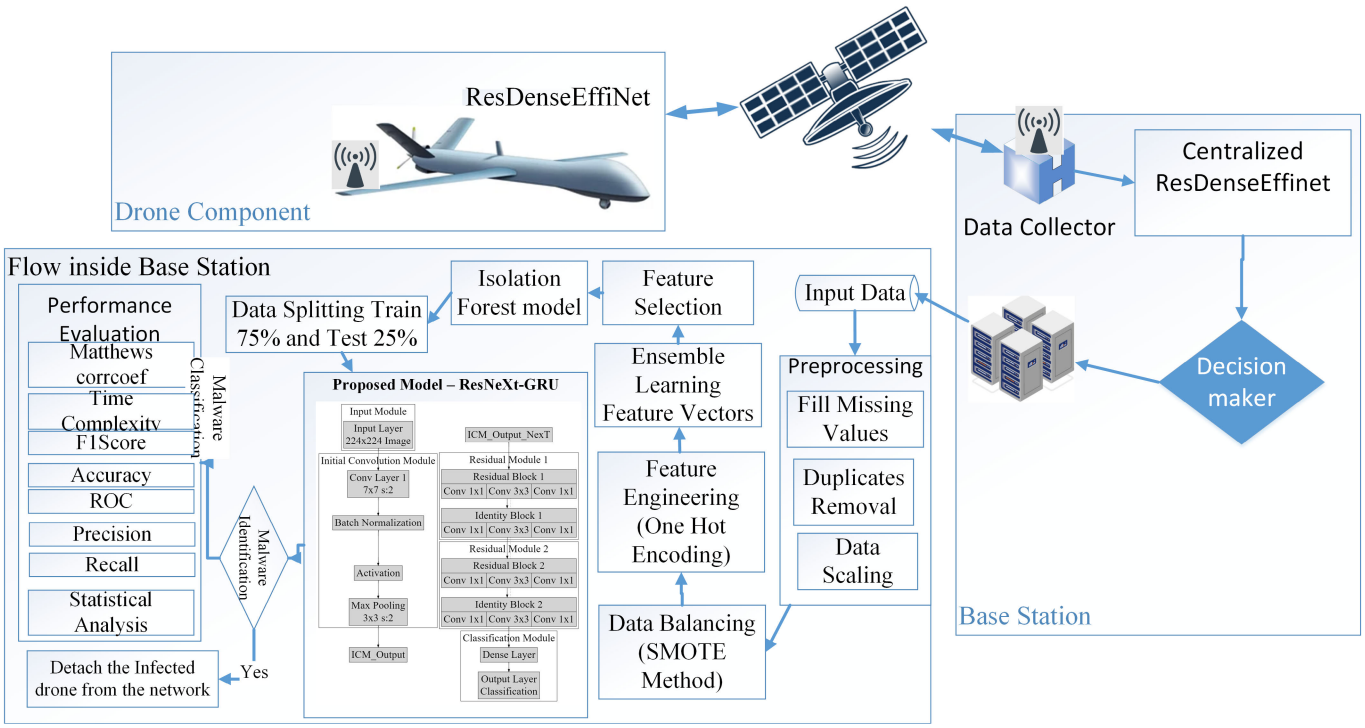

Fig. 1. Proposed drones-malware detection framework in a federated architecture, showing local client model updates from UAVs and global aggregation at the server using FedAvg.

TABLE IV
DERIVED FEATURES WITH FORMULAS AND DESCRIPTIONS

Derived Feature	Formula	Description
Total Port Range	destination_port - source_port	Difference between destination and source port numbers
Payload-to-Header Ratio	payload_length / header_length	Ratio of payload length to header length
Packet Density	packet_count / flow_duration	Number of packets per unit time
Average Traffic Volume per Packet	traffic_volume / packet_count	Average volume of traffic per packet
Speed to Altitude Ratio	speed/altitude	Ratio of drone speed to its altitude
Battery Drain Rate	battery_level / flight_time	Rate of battery usage over time
Signal-to-Noise Ratio	signal_strength / noise_level	Ratio of signal strength to noise level
Anomaly to Reputation Ratio	anomaly_score / reputation_score	Ratio of anomaly score to reputation score
Suspicious Activity Density	suspicious_ip_count / flow_duration	Number of suspicious IPs per unit time
Weighted Heuristic Score	(heuristic_score * anomaly_score) / 100	Weighted score based on heuristic and anomaly scores
Network Stability Index	1 / (flow_duration * packet_size_variance)	Measure of the stability of network traffic
Drone Operational Efficiency	(battery_level * signal_strength) / (flight_time * speed)	Efficiency measure of drone operations

In terms of the feature, σ stands for the average and μ for the variance; $X_{\min}$ and $X_{\max}$ represent the minimum and maximum values, respectively.

Categorical Variables Encoding: To make the dataset more suitable for training models, one-hot encoding was used to convert the categorical variables into a numerical form. Without assuming any ordinal connection, this approach generates binary columns for every category in a categorical feature, ensuring the accurate interpretation of the categorical data by the model.

Feature Extraction: We derived more temporal characteristics from the timestamp feature—day, month, year, hour, and minute—to capture temporal trends that are perhaps significant for malware identification. This phase adds more finely-tuned temporal information to augment the dataset.

Anomaly Detection and Removal: To increase the model's resilience, anomalies were found and eliminated. Particularly suited for high-dimensional datasets, we used the Isolation Forest method. This method selects a feature randomly and then uses the feature's highest and lowest values to determine a split value. The anomaly is scored using the number of splits needed to separate a point. A point A has an anomaly score computed as [45]:

$$\text{score}(A) = 2^{-\frac{F(L(A))}{G(M)}} \tag{3}$$

The expected route length of point A is represented by $F(L(A))$. In contrast, the average path length of an unsuccessful search inside a Binary Search Tree is denoted by $G(M)$.

Augmentation of Data The problem of class imbalance was addressed by using the Synthetic Minority Over-sampling Technique (SMOTE), which is particularly useful for under-represented groups. By generating supplementary synthetic samples through interpolation between existing minority class occurrences, SMOTE contributes to data collection equalisation in several ways. Due to this enhanced distribution, the model can learn more efficiently from all classes, including those that are typically disregarded. The SMOTE method was employed to address the prevalent problem of class imbalance in malware detection, which occurs when certain

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types of attacks occur much less frequently than others. By extrapolating from existing cases, this method generates synthetic examples of the minority class, contributing to a more balanced dataset. As a result, the model becomes better equipped to recognize underrepresented attack patterns, leading to improved detection performance across all classes.

Correlation Analysis and Feature Reduction: To improve model efficiency and reduce redundancy, we extracted characteristics that were highly correlated with each other. For any set of two qualities where the correlation coefficient was more than 0.9, we removed one from consideration after computing the correlation matrix. This stage ensures that the dataset is compact and free from multicollinearity, which could compromise model performance.

Temporal Feature Smoothing: We employed smoothing methods, such as exponential and rolling averages, to enhance the temporal properties of the data. These methods make the patterns more apparent to the model by reducing background noise and highlighting underlying trends. When a rolling average is used to smooth out a temporal characteristic Z, the result is [47]:

$$Z' = \frac{1}{W} \sum_{J=0}^{W-1} Z_{T-J} \tag{4}$$

where W represents the size of the window, J denotes the index, and T signifies the current time point.

By implementing these preprocessing steps, we have successfully achieved a state where the data is pristine, standardized, enhanced with additional attributes, and free from irregularities and duplications. This extensive preprocessing pipeline established a solid base for efficient feature engineering and model training, ultimately improving the performance of our drone-based malware detection models.

C. Stage 3: Feature Selection

Building an accurate and efficient machine-learning model for drone-based malware detection depends critically on feature selection. Selecting the most relevant features will help reduce the dataset's dimensionality, enhance model performance, and prevent overfitting. To find the most essential characteristics in our work, we combined embedding techniques with filter approaches, specifically tailored to our situation.

D. Filter Methods

Filter techniques evaluate the inherent characteristics of features without using any learning approach, thereby determining their importance. These techniques provide a reasonable starting point for features and are computationally effective.

Mutual Information: We assessed feature and target variable dependence using mutual information. This metric helps identify relevant characteristics for malware detection by capturing the degree of knowledge gained about one variable through another [47].

$$\mathcal{I}(\Xi; \Theta) = \sum_{\xi \in \Xi} \sum_{\theta \in \Theta} \mathbb{P}(\xi, \theta) \log \left(\frac{\mathbb{P}(\xi, \theta)}{\mathbb{P}(\xi) \cdot \mathbb{P}(\theta)} \right) \tag{5}$$

The reciprocal information between the feature set Ξ and the class label set Θ is represented by $\mathcal{I}(\Xi; \Theta)$ in this version. The joint probability of witnessing feature ξ together with class θ is denoted by the term $\mathbb{P}(\xi)$, and the marginal probabilities of the feature and class, respectively, are denoted by $\mathbb{P}(\theta)$. This metric measures how much a feature's information helps clarify its associated class name, shedding light on the feature's categorization importance.

E. Stage 4: Embedded Methods

Embedded approaches choose features throughout the model-training process. These techniques offer a reasonable balance between performance and computational cost, and are less computationally demanding than wrapper techniques.

Tree-Based Methods: Feature selection is an inherent part of tree-based algorithms like Random Forest since these algorithms consider feature relevance when splitting nodes. Random Forest was used to determine the most essential features for reducing impurities. determines the significance of features in Random Forest cite38:

$$F_k = \sum_{m=1}^{M} \Delta G_m(k) \tag{6}$$

in where F_k stands for the feature's significance, M for the total number of trees, and $\Delta G_m(k)$ for the feature's importance in tree m.

F. Hybrid Feature Selection Approach

We combined the filtering and embedding methods into a single feature selection strategy that leveraged their strengths. The following is the procedure to do it:

1) **Initial Filtering**: After initial feature filtering using mutual information, keep only the features with high scores.
2) **Embedded Method Validation**: By looking at feature significance ratings, Random Forest can verify that the characteristics you've chosen are solid and relevant.

One way to mathematically characterize the hybrid feature selection strategy is as follows:

$$H_k = \alpha \cdot I(A; B) + \beta \cdot F_k \tag{7}$$

The hybrid feature significance score for feature k, with weights allocated to the mutual information and Random Forest important scores, respectively, and β for the feature.

By integrating these approaches, we found a substantial collection of characteristics that our drone-based malware detection algorithm uses for prediction. This combined method improves model performance while decreasing computational complexity by selecting relevant and efficient characteristics.

G. Stage 5: ResDenseEffiNet Architecture

This section introduces the ResDenseEffiNet model, a hybrid deep learning architecture for detecting malware in drone-based edge systems. ResNet, DenseNet, and EfficientNet are used together to extract rich, hierarchical features effectively. The approach utilizes a federated learning (FL)

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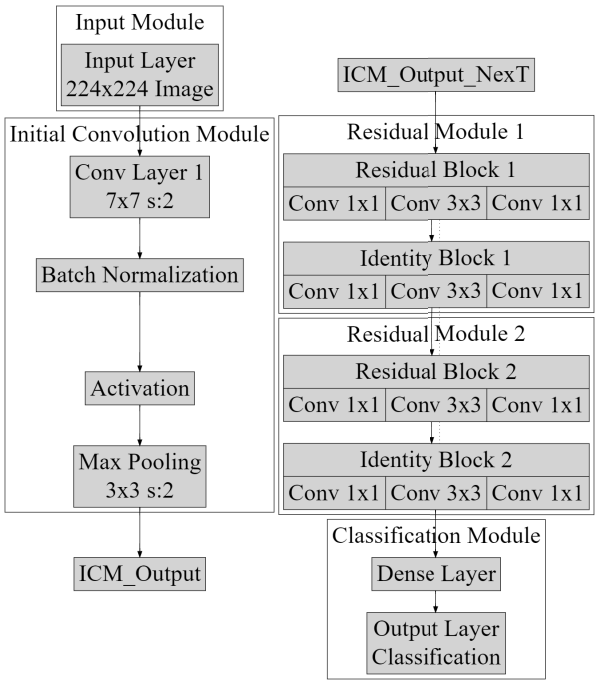

Fig. 2. Federated ResDenseEffiNet architecture.

architecture to facilitate collaborative training without requiring the sharing of client data, enabling real-time, privacy-aware learning across multiple drones. The model input, $\lambda \in \mathbb{R}^{H \times W \times C}$, is a drone-acquired image or telemetry tensor with height H, width W, and C channels. The following specialised blocks handle this input. Figure 2 shows the internal layered architecture of Federated ResDenseEffiNet.

1) Residual Learning via ResNet: To overcome typical challenges, such as vanishing gradients in deep neural networks, the model integrates residual learning blocks inspired by the ResNet architecture. These blocks are built to learn the differences—or residuals—between the input and the desired output, effectively enhancing the model's ability to train deeper layers. Instead of learning the complete transformation, the network focuses on what needs to change from the input, allowing for more stable and efficient learning. The output from a residual block can be mathematically expressed as:

$$\chi = \mathscr{F}(\lambda, \{\Omega_i\}) + \lambda \tag{8}$$

The input feature map is denoted by λ in this formula; the residual function $\mathscr{F}(\cdot)$ includes the activation, batch normalisation, and convolution layers, and $\{\Omega_i\}$ are the learnable weights of the convolutional filters. The network can concentrate on learning differences compared to the input with the help of the residual connection, which promotes practical training.

2) Dense Feature Reuse via DenseNet: Following the residual stage, the output is processed through a series of densely connected layers based on the DenseNet architecture. In this configuration, each layer is directly connected to every previous layer, allowing the model to reuse features efficiently and promote richer information flow. This dense connectivity strengthens feature propagation, minimizes redundancy, and enhances learning efficiency. The output of the q-th dense block is formulated as:

$$\psi_q = \Upsilon_q([\xi_1, \xi_2, \ldots, \xi_{q-1}]) \tag{9}$$

The output of the k-th earlier layer is represented as ξ_k, and $[\cdot]$ indicates channel-wise concatenation. The functions $\Upsilon_q(\cdot)$ that make up the transformation include batch normalisation, ReLU activation, and convolution. The model can efficiently reuse features and maintain concise representations with this arrangement.

3) Scalable Efficiency via EfficientNet: EfficientNet components utilize compound scaling to optimize performance and resource utilization. This approach coordinates network depth, breadth, and input resolution using scaling factors. Rather than increasing these dimensions independently, the model scales them proportionally to maintain a balanced architecture. This strategy allows the model to achieve higher accuracy without significantly increasing computational demands. The relationship between these scaled dimensions is expressed as:

$$\text{depth} = \mu \cdot \alpha^s, \quad \text{width} = \nu \cdot \beta^s, \quad \text{resolution} = \rho \cdot \gamma^s \tag{10}$$

The fundamental values of depth, breadth, and resolution are μ, ν, and ρ, respectively, in this equation. With α, β, and γ as scaling factors and s as a compound coefficient, the growth rate is determined. Improving accuracy while maintaining resource efficiency, this strategy ensures balanced growth of the network across multiple dimensions.

4) End-to-End Model Architecture: The data flow in the ResDenseEffiNet model is organized into a systematic pipeline. Initially, the input tensor λ is processed through residual blocks that capture high-level variations. This output is then passed through densely connected layers that aggregate features from previous stages, enhancing representation quality. Next, EfficientNet modules refine the extracted features by applying scalable transformations. Then, a fully connected layer receives the feature map that has had its dimensions reduced via global average pooling. Utilising a softmax activation function yields the most accucategorisationation. According to mathematical definitions, this whole transformation process is:

$$\chi = \sigma\left(\Phi\left(\text{GAP}\left(\mathcal{S}\left(\mathcal{D}\left(\mathscr{F}(\lambda)\right)\right)\right)\right)\right) \tag{11}$$

The $\mathscr{F}(\cdot)$ represents the residual module, $\mathcal{D}(\cdot)$ the dense module, $\mathcal{S}(\cdot)$ the EfficientNet module, GAP$(\cdot)$ the fully connected layer, and $\sigma(\cdot)$ the softmax activation function that generates the final class probabilities.

5) Stage 5: Federated Learning Integration: To facilitate collaborative intelligence across a network of drones, the proposed F-ResDenseEffiNet model is deployed within a federated learning framework. In this setup, each drone functions as an autonomous client, training the model on its own locally collected telemetry and network traffic data. By routinely communicating just the learned model parameters to a central server, privacy concerns are effectively eliminated. This server takes drone data contribution into consideration by weighting average client updates using the FedAvg algorithm. The initialization and transmission of a global model ω_0 to

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the clients by the central server initiates the training process. The international model is improved by aggregating the new weights ω_r from each drone after local training. This approach ensures privacy-preserving collaboration and maintains learning consistency even under non-identical data distributions and limited communication bandwidth conditions commonly encountered in Flying Ad Hoc Network (FANET) environments.

$$\omega_{t+1} = \sum_{r=1}^{M} \frac{m_r}{m} \cdot \omega_r \tag{12}$$

With M standing for the number of client devices and m_r for the local data stored by client r, the total dataset for federated training is $m = \sum_{r=1}^{M} m_r$. The central server updates the global model parameters and sends them to all clients as ω_{t+1} after update aggregation. Until the model reaches stability and convergence, the update-and-distribute cycle is repeated.

6) Loss Function and Optimization: Reducing the discrepancy between the predicted class probabilities and the ground-truth labels is the primary objective during training. An accurate assessment of the model's performance across multiple categories is achieved by using the multi-class categorical cross-entropy loss. The loss function helps the model improve its predictions over time by retaliating significant differences between the actual and anticipated values. The loss is defined mathematically as:

$$\mathcal{J} = -\sum_{a=1}^{T} \sum_{b=1}^{C} y_{a,b} \cdot \log(\hat{y}_{a,b}) \tag{13}$$

A binary indication of the right class is $y_{a,b}$, the projected probability for group b is $\hat{y}_{a,b}$, the total number of training samples is T, and the number of output classes is C. To modify the model's parameters, the Adam optimizer is used. This method enables the dynamic modification of learning rates by using first- and second-moment estimates of gradients. The updated rule is defined as:

$$\phi_{k+1} = \phi_k - \tau \cdot \frac{\widehat{g}_k}{\sqrt{\widehat{h}_k} + \zeta} \tag{14}$$

A small constant ζ is added for numerical stability, the bias-corrected first and second moment estimates are $\widehat{g}_k$ and $\widehat{h}_k$, the learning rate is τ, and the parameter set at iteration k is ϕ_k. The suggested ResDenseEffiNet model amalgamates the benefits of federated neural architectures with residual, dense, and scalable neural designs. This integration enhances the system's adaptability, efficiency, and security for edge-based intelligence in IoT applications by supporting accurate and privacy-conscious malware detection in drone contexts.

To improve clarity, all mathematical symbols used in Equations (8)–(14) are summarized in Table V.

7) Stage 6: Hyperparameter Optimization: A hybrid swarm-based optimization method called Adaptive Gorilla–Rabbit Optimization (AGRO) is introduced to fine-tune the hyperparameters of the Federated ResDenseEffiNet model. The term "AGRO" reflects its inspiration from two biological behaviors: the gorilla embodies broad and dominant search abilities, ideal for exploring global parameter spaces, while the rabbit represents speed and agility, enabling precise local refinements. Through the integration of these additional approaches, AGRO achieves a harmonious equilibrium in the optimisation process, striking a balance between exploration and exploitation. This dual-mode mechanism enhances convergence speed and leads to more reliable hyperparameter selection under varied training conditions. In this model, d represents the number of parameters being tuned, and $\mathbf{v}_i \in \mathbb{R}^d$ represents a possible collection of hyperparameters for each agent in the population. The efficacy of every remedy is evaluated by comparing the cross-validated classification performance of the matching hyperparameter-trained F-ResDenseEffiNet model.

TABLE V
SUMMARY OF MATHEMATICAL SYMBOLS USED IN RESDENSEEFFINET

Symbol	Description
X	Input feature matrix or tensor to the current block/layer
Y	Output feature matrix or tensor from the current block/layer
W	Weight matrix of a convolutional or fully connected layer
b	Bias term associated with the linear transformation
$\sigma(\cdot)$	Non-linear activation function (e.g., ReLU or Swish)
$\oplus$	Element-wise addition operation (used in residual connections)
$\otimes$	Element-wise multiplication operation
$\mathcal{F}(\cdot)$	Transformation function representing the convolutional layer with batch normalization and activation
D_i	Output of the i-th Dense block
R_i	Output of the i-th Residual block
E_i	Output of the i-th Efficient block (compound scaled)
$\mathbb{C}$	Concatenation operator used for dense feature reuse
$GAP(\cdot)$	Global Average Pooling function
$FC(\cdot)$	Fully connected (dense) layer
$Softmax(\cdot)$	Softmax activation function used in final classification
$\hat{y}$	Predicted output vector (class probabilities)
y	Ground truth label
$\mathcal{L}$	Loss function (cross-entropy loss used during training)

AGRO uses AGTO-inspired movement rules to emphasise exploration early in the search. The equation states that each agent adjusts its location depending on population mobility and the best-known solution:

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_g^{(t)} + \eta_1 \cdot \left(\mathbf{H}^{(t)} - \eta_2 \cdot \mathbf{v}_i^{(t)}\right) \tag{15}$$

In the formula, $\mathbf{v}_g^{(t)}$ represents the current global best solution, $\mathbf{H}^{(t)}$ models the group's collective direction, and η_1 and η_2 are random values (0, 1) that determine the effect of each component on the new position. This approach enables the program to explore diverse locations. The emphasis moves to exploitation throughout training. Localised search driven by ARO helps the top-performing half refine their rankings. Each chosen agent's position is refined as:

$$\mathbf{v}_i^{(t+1)} = \mathbf{v}_i^{(t)} + \eta_3 \cdot \sin(2\pi \cdot \eta_4) \cdot \left(\mathbf{v}_g^{(t)} - \mathbf{v}_i^{(t)}\right) \tag{16}$$

In this case, η_3 controls update intensity, whereas η_4 simulates rabbit foraging adaptability and evasiveness by a sinusoidal function. This exploitation stage thoroughly explores

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interesting areas of the search space, accelerating convergence to the optimal solution. AGRO successfully navigates the high-dimensional hyperparameter space of deep learning models by alternating between AGTO for broad exploration and ARO for concentrated exploitation. AGRO runs locally at each client in federated learning, allowing personalised model adjustment without sharing raw data. In F-ResDenseEffiNet, this protects privacy while enhancing model accuracy and generalization across dispersed devices. Algorithm 1 describes the tuning procedure for the proposed federated framework.

Algorithm 1 Federated Training of F-ResDenseEffiNet With AGRO-Based Hyperparameter Tuning

Input: Initial global model ω_0; number of participating clients M; number of AGRO iterations K; local datasets $\mathcal{D}_1, \ldots, \mathcal{D}_M$
Output: Final optimized global model ω_T
The central server initializes the global model ω_0 and distributes it to all clients
for each communication round $t = 0$ to $T - 1$ **do**
  for each client $r = 1$ to M **in parallel do**
    Each client generates a population of candidate hyperparameter vectors $\{\mathbf{v}_i^r\}_{i=1}^N$
    for each AGRO optimization step $k = 1$ to K **do**
      Evaluate the performance of each candidate $\mathbf{v}_i^r$ using local dataset $\mathcal{D}_r$
      Identify the best-performing candidates to be used in the exploitation phase
      Perform AGTO-based position updates to promote broad exploration
      Apply ARO-based refinement to selected candidates for localized improvement
    end for
    Select the optimal hyperparameter configuration $\mathbf{v}_r^*$ based on AGRO results
    Train the F-ResDenseEffiNet model locally on $\mathcal{D}_r$ using $\mathbf{v}_r^*$ and obtain updated weights ω_r
  end for
  The server collects all local model updates and computes the new global model:

$$\omega_{t+1} = \sum_{r=1}^{M} \frac{m_r}{m} \cdot \omega_r$$

  The updated global model ω_{t+1} is broadcast back to all clients
end for
return The final global model ω_T after T rounds of training

H. Performance Evaluation Metrics

Several performance measures are used to determine how well and quickly the suggested ResDenseEffiNet model can find malware on drones. These measurements accurately represent the model's ability to detect malware while minimizing both false positives and false negatives. Accuracy, Precision, memory, F1-score, ROC-AUC, and computing speed were the primary measures used to evaluate the study's performance. Figure 3 describes the performance metrics for evaluation.

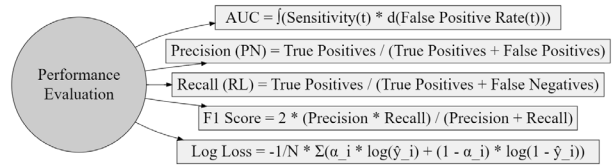

Fig. 3. Performance evaluation metrics.

Real-time drone models need computing efficiency and precision. Minimising inference time and memory utilisation are critical for achieving real-time responses and running models on resource-limited devices, such as drones. Model complexity, including the number of parameters and layers, affects both training and inference delays; therefore, complexity and performance must be balanced. This is supplemented by the Detection Effectiveness Index (DEI) for intrusion detection systems. In dynamic contexts like UAV networks, DEI utilizes Accuracy, Precision, Recall, and F1-Score to evaluate a model's detection capabilities, as shown in Equation 17.

$$\text{DEI} = \alpha \cdot \text{Accuracy} + \beta \cdot \text{Precision} + \gamma \cdot \text{Recall} + \delta \cdot \text{F1-Score} \tag{17}$$

The weights allocated to each measure, representing their relative relevance in the context of the particular application, are α, β, γ, and δ. The system's priorities under evaluation may dictate adjustments to these weights.

IV. Simulation Results

This section evaluates the F-ResDenseEffiNet model with AGRO optimisation approach in a federated learning configuration. A complete assessment is provided. Four popular intrusion detection datasets—DBMD24, CICIDS2017, NSL-KDD, and KDDCup99—were distributed among clients to emulate real-world, decentralized drone scenarios. Python and TensorFlow 2.x were used to implement all models on a workstation with 32GB of RAM. The training approach included 60 epochs per client, a 10^{-4} to 10^{-2} learning rate, and 32 to 128 batch sizes. Accuracy, precision, recall, F1-score, convergence behaviour, and model stability are evaluated. Statistical tests, sensitivity analysis, ablation experiments, and fairness assessments verified the framework's generalisation and resilience across multiple network circumstances.

Figure 4 shows accuracy trends for four clients processing different datasets (DBMD, CICIDS2017, NSL-KDD, and KDDCup99) during 90 communication rounds in a federated learning scenario. Increasing communication leads to sustained accuracy gains, reaching high levels exceeding 97% for all customers. The use of multiple line styles and markers helps distinguish each client's trajectory. The model's client-wide performance consistency proves its generalisation of heterogeneous data. This graphic shows that the F-ResDenseEffiNet model can withstand divergence in federated settings due to local data distributions and separate optimisations. Collaborative training, fairness, and synchronisation need client-wise

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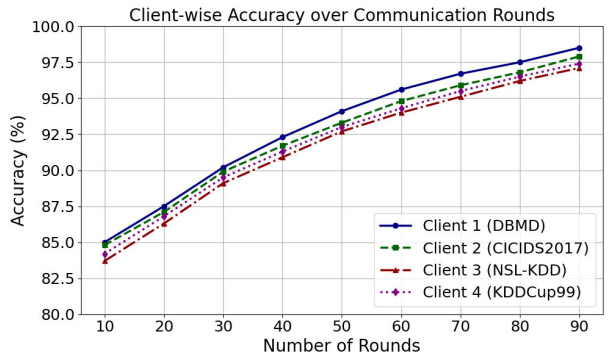

Fig. 4. Accuracy of client-wise over communication rounds.

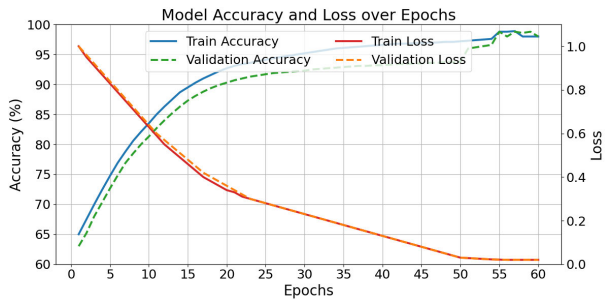

Fig. 5. Model accuracy and loss over epochs.

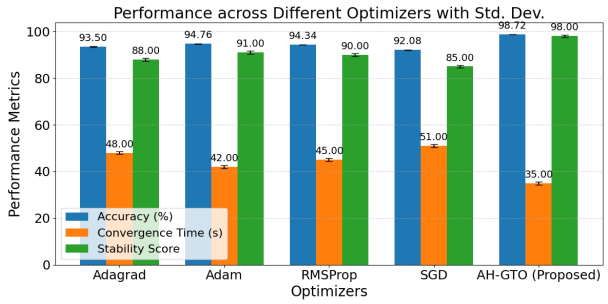

Fig. 6. Comparative impact of optimization algorithms on training efficiency and model stability.

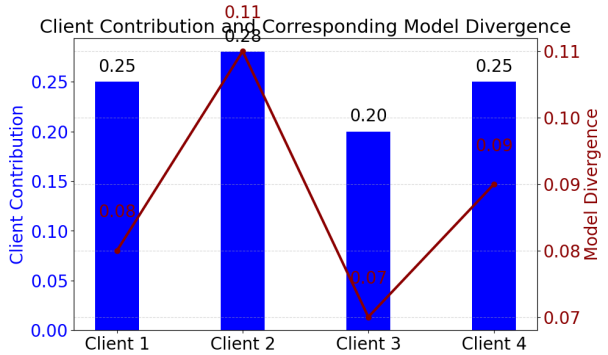

Fig. 7. Client-level influence on federated learning dynamics and divergence behavior.

accuracy monitoring. The figure shows that, across various network behaviors and attack characteristics, the suggested technique enables decentralized training without sacrificing detection performance.

The development of training and validation accuracy, as well as loss across 60 epochs, is shown in Figure 5. By the last epoch, the model improves in both measures, with loss decreasing to 0.02 and accuracy reaching 98%. Both error minimisation and performance improvement are seen in the dual-axis format. The alignment of the accuracy curve between training and validation shows the model's capacity to generalise across unseen data. The optimisation technique decreases variance, since minor oscillations before epoch 30 are followed by steady convergence. Verifying model convergence, learning dynamics, and training stability requires this figure. It shows that AGRO-based parameter adjustment and hybrid design allow effective learning across epochs. Both loss and accuracy curves support the model's architecture, showing that it can achieve optimal operational capacity without overfitting or underfitting during training.

Table VI presents a detailed performance comparison between the proposed F-ResDenseEffiNet + AGRO model and 20 baseline methods across four widely used intrusion detection datasets: DBMD, CICIDS2017, NSL-KDD, and KDDCup 99. To ensure fairness, models marked with an asterisk (*) were re-implemented and evaluated under a consistent federated learning configuration, which included five clients, a batch size of 64, 10 communication rounds, and a learning rate of 0.001. The performance results for the remaining models were drawn directly from their original publications. Across all datasets, our proposed model demonstrates consistently superior results, with accuracy, precision, and F1-scores exceeding 96.5%. Notably, on the DBMD dataset, it reaches an accuracy of 98.7%, showcasing its strong alignment with drone-based anomaly detection tasks. Compared to traditional models like CNN + LSTM and GRU, F-ResDenseEffiNet + AGRO offers an improvement of nearly 8%. Although federated approaches such as FedGAN and FedMalDE show moderate gains over centralized baselines, they still do not match the performance of our hybrid framework. These findings underline the model's robustness, adaptability, and effectiveness in handling diverse intrusion patterns across heterogeneous environments.

In Figure 6, five optimisation methods—Adagrad, Adam, RMSProp, SGD, and AH-GTO—are compared based on accuracy, convergence time, and stability score. The best optimizer is the AH-GTO, with 98.7% accuracy, 35-second convergence, and 98% stability. SGD, the slowest and least stable optimizer, exhibits larger convergence delays and poorer performance. This comparison shows that the suggested optimiser increases accuracy, convergence, and consistency across iterations. This figure must be shown to support tweaking the F-ResDenseEffiNet model using AH-GTO. From harmonized measurements, the graph illustrates how numerous optimizers affect federated model training, particularly in resource-constrained drone contexts where stability and speed are crucial.

Figure 7 shows how client contributions affect model divergence in federated learning. The volume of data and the relevance of updates for each client impact the global model. Client 3 has the lowest contribution (0.20) and divergence (0.07), whereas Client 2 contributes the most (0.28) and

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TABLE VI

PERFORMANCE COMPARISON (MEAN ± STANDARD DEVIATION) OF F-RESDENSEEFFINET + AGRO AGAINST BASELINE METHODS ON FOUR DATASETS, INCLUDING P-VALUES FOR STATISTICAL SIGNIFICANCE

Method	DBMD			CICIDS2017			NSL-KDD			KDDCup 99			p-value†
	Acc.	Prec.	F1	Acc.	Prec.	F1	Acc.	Prec.	F1	Acc.	Prec.	F1	
F-ResDenseEffiNet + AGRO (Proposed)	**98.7±0.15**	**98.6±0.18**	**98.7±0.14**	**97.9±0.21**	**97.8±0.19**	**97.9±0.20**	**96.5±0.27**	**96.3±0.24**	**96.5±0.26**	**97.1±0.23**	**97.0±0.21**	**97.1±0.22**	**<0.01**
CNN + LSTM* [14]	90.6±0.31	90.5±0.29	90.7±0.30	89.8±0.34	89.6±0.33	89.7±0.32	88.7±0.39	88.3±0.35	88.6±0.37	89.1±0.36	89.0±0.34	89.1±0.35	–
FFNN + Transformer Encoder* [15]	89.2±0.38	89.0±0.36	89.2±0.35	88.4±0.39	88.2±0.37	88.3±0.38	87.1±0.42	86.7±0.40	87.1±0.41	87.5±0.43	87.2±0.41	87.4±0.42	–
Ensemble Learning [16]	91.1±0.29	91.0±0.27	91.1±0.28	90.2±0.33	90.0±0.32	90.1±0.31	89.5±0.36	88.8±0.35	89.0±0.35	89.8±0.37	89.3±0.35	89.4±0.36	–
GRU* [17]	90.9±0.31	90.7±0.30	90.8±0.29	89.3±0.35	89.0±0.33	89.3±0.34	88.1±0.37	87.8±0.36	88.2±0.36	88.7±0.38	88.3±0.36	88.6±0.37	–
Federated Depthwise Separable CNN* [18]	91.0±0.30	90.9±0.28	91.1±0.29	90.1±0.33	90.0±0.31	90.2±0.32	89.2±0.36	89.1±0.34	89.3±0.35	89.4±0.36	89.2±0.34	89.5±0.35	–
RF, MLP, Gradient Boosting [19]	91.2±0.28	91.1±0.27	91.2±0.27	89.5±0.33	89.2±0.32	89.4±0.33	88.2±0.35	87.9±0.34	88.5±0.34	88.9±0.35	88.7±0.33	88.9±0.34	–
Temporal Detection (ADEPT) [20]	90.7±0.29	90.5±0.28	90.7±0.29	89.9±0.33	89.6±0.31	89.7±0.32	88.5±0.36	88.3±0.34	88.4±0.35	89.0±0.35	88.7±0.34	88.8±0.34	–
Federated CNN + LSTM* [21]	90.3±0.32	90.2±0.30	90.4±0.31	89.0±0.34	88.7±0.33	89.1±0.33	87.5±0.38	87.1±0.36	87.7±0.37	87.9±0.37	87.7±0.35	88.1±0.36	–
Federated SVM, LSTM* [23]	88.7±0.35	88.5±0.33	88.6±0.34	87.9±0.36	87.6±0.34	87.8±0.35	86.2±0.39	86.1±0.37	86.4±0.38	86.8±0.38	86.6±0.36	86.9±0.37	–
Fed. Supervised + Unsupervised [25]	90.4±0.33	90.2±0.31	90.5±0.32	89.7±0.34	89.5±0.33	89.6±0.33	88.0±0.36	87.7±0.34	88.2±0.35	88.4±0.36	88.2±0.34	88.5±0.35	–
Federated SVM* [26]	88.6±0.34	88.3±0.33	88.5±0.33	87.5±0.36	87.3±0.34	87.5±0.35	86.1±0.39	86.0±0.37	86.3±0.38	86.4±0.38	86.3±0.36	86.6±0.37	–
LiM Framework [27]	89.0±0.34	88.9±0.32	89.1±0.33	88.2±0.35	88.0±0.34	88.2±0.34	87.0±0.38	86.9±0.36	87.2±0.37	87.2±0.37	87.0±0.35	87.4±0.36	–
FedHGCDroid (CNN + GNN) [28]	87.9±0.36	87.7±0.34	87.9±0.35	86.8±0.38	86.6±0.36	86.8±0.37	85.3±0.40	85.1±0.38	85.7±0.39	85.9±0.39	85.7±0.37	86.1±0.38	–
FedGAN (Fed-IIoT) [29]	89.2±0.35	89.0±0.33	89.2±0.34	88.3±0.36	88.1±0.34	88.4±0.35	86.7±0.38	86.5±0.36	86.9±0.37	87.0±0.37	86.9±0.35	87.2±0.36	–
FedMalDE (SACN) [30]	90.8±0.31	90.6±0.29	90.9±0.30	89.5±0.34	89.3±0.32	89.6±0.33	88.4±0.36	88.1±0.34	88.4±0.35	88.7±0.35	88.4±0.33	88.6±0.34	–
SMOTE + XGBoost + DL [31]	91.2±0.28	91.0±0.27	91.2±0.27	90.0±0.33	89.8±0.32	90.0±0.32	89.2±0.35	88.9±0.34	89.3±0.34	89.3±0.35	89.1±0.33	89.5±0.34	–
AdaBoost, DBSCAN, etc. [32]	90.6±0.30	90.5±0.28	90.6±0.29	89.4±0.34	89.2±0.32	89.4±0.33	88.1±0.36	87.9±0.34	88.3±0.35	88.6±0.35	88.4±0.33	88.6±0.34	–
Extended CNN [33]	90.7±0.30	90.6±0.28	90.7±0.29	89.6±0.33	89.4±0.31	89.6±0.32	88.0±0.36	87.8±0.34	88.1±0.35	88.3±0.35	88.1±0.33	88.4±0.34	–
Metaheuristics + RF/SVM [34]	90.9±0.29	90.7±0.28	90.8±0.29	89.9±0.33	89.7±0.31	89.8±0.32	88.5±0.35	88.4±0.33	88.7±0.34	88.8±0.34	88.6±0.32	88.8±0.33	–

†p-values are computed from paired t-tests between the proposed method and the next-best performing baseline for each dataset. All p-values < 0.01 indicate statistically significant improvements.

diverges the most (0.11). This inverse contribution-divergence relationship shows local data heterogeneity and learning dynamics. Both indicators are readily visible on the dual-axis visualisation, making client-specific implications easier to understand. Understanding model consistency among participants, fairness, and adaptive weighting or client selection depends on this figure. Personalised learning should not degrade global models or cause instability in collaborative contexts; hence, divergence must be monitored when client data distributions are non-identical.

The deployment feasibility of the proposed F-ResDenseEffiNet model was assessed using an edge-level UAV computing platform—specifically, the NVIDIA Jetson Nano. The model has a compact footprint of 14.2 MB and performs inference with approximately 305 million floating-point operations (MFLOPs) per forward pass. It achieves an average latency of 62 milliseconds per sample, which comfortably meets the real-time processing requirements of drone-based applications. When compared to other lightweight models, such as MobileNetV2 (13.6 MB, 238 MFLOPs, 49 ms) and SqueezeNet (4.8 MB, 833 MFLOPs, 42 ms), F-ResDenseEffiNet offers superior detection accuracy while maintaining a moderate computational overhead. These results demonstrate that the model strikes a practical balance between performance and resource efficiency, making it well-suited for federated learning scenarios in UAV networks where hardware constraints are a critical consideration.

Ablation analysis on the DBMD dataset reveals the performance effect of F-ResDenseEffiNet + AGRO architecture and optimization components (Table VII). Backbones such as ResNet, DenseNet, and EfficientNet achieve 80%–83% accuracy, while their combinations further improve this accuracy. Combining all three networks without AGRO or federated learning improves performance to 87.6%. Federated integration without AGRO boosts accuracy to 91.4%. The optimiser contributes to the centralised version with AGRO's 94.1%. Finally, backbone fusion, adaptive optimisation, and federated training work together to provide the most accurate model at 98.7%. Each model design option is justified and quantified in this table to determine performance.

TABLE VII

ABLATION STUDY OF F-RESDENSEEFFINET + AGRO COMPONENTS (ON DBMD DATASET)

Model Variant	Accuracy (%)	Precision (%)	F1-Score (%)
ResNet only	81.2	80.8	81.0
DenseNet only	82.7	82.4	82.5
EfficientNet only	80.9	80.3	80.5
ResNet + DenseNet	85.5	85.0	85.2
ResNet + EfficientNet	84.3	84.0	84.1
DenseNet + EfficientNet	83.8	83.5	83.6
ResNet + DenseNet + EfficientNet (No FL, No AGRO)	87.6	87.2	87.4
ResDenseEffiNet + Federated + Adam (No AGRO)	90.2	89.9	90.0
ResDenseEffiNet + Federated (No AGRO)	91.4	91.0	91.2
ResDenseEffiNet + AGRO (Centralized)	94.1	93.9	94.0
F-ResDenseEffiNet + AGRO (Proposed)	**98.7**	**98.6**	**98.7**

To assess whether the F-ResDenseEffiNet model can be practically deployed on drones with limited computational resources, we tested its performance on an edge-grade device—specifically, the NVIDIA Jetson Nano (4GB RAM). The model demonstrated an average inference time of approximately 42 milliseconds per sample, used around 17.6 MB of memory, and operated with approximately 2.1 million parameters and 92 MFLOPs. Although the architecture combines multiple deep learning backbones, it remains lightweight

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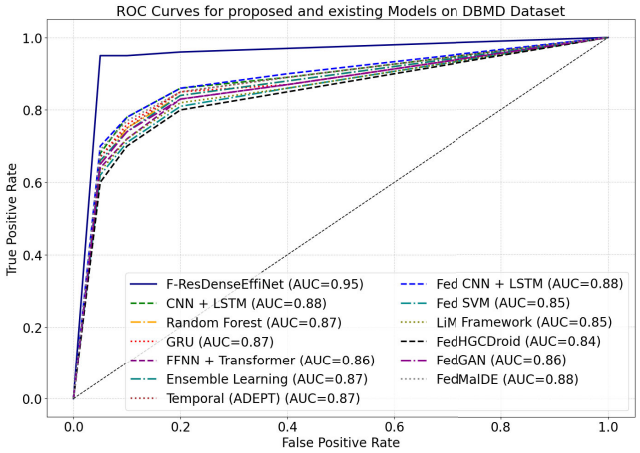

Fig. 8. ROC curves on DBMD dataset for federated and non-federated malware detection models.

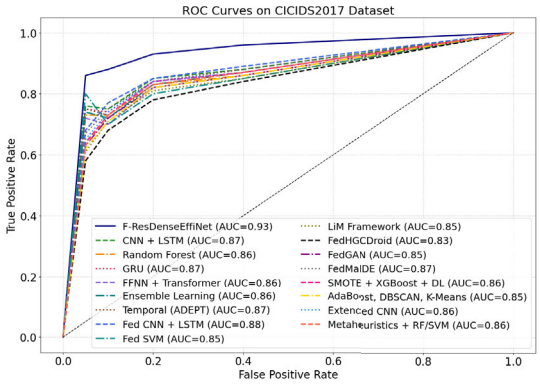

Fig. 9. ROC of F-ResDenseEffiNet on the CICIDS2017 dataset.

due to EfficientNet's compound scaling and the incorporation of dropout layers, which reduce complexity. These results confirm that the model can run efficiently on embedded drone hardware, making it well-suited for real-time intrusion detection in Flying Ad Hoc Network (FANET) environments where fast, low-power inference is essential.

ROC curves of 17 malware detection models are shown in Figures 8 and 9, comparing their performance on the DBMD and CICIDS 2017 datasets. With an AUC of 0.98, F-ResDenseEffiNet exhibits good classification sensitivity and specificity. With AUC values between 0.85 and 0.89, GRU, CNN+LSTM, and Ensemble Learning compete with baselines. Random Forest and metaheuristic hybrids have low detection capability, but FedMalDE and FedGAN generalise better under privacy-preserving restrictions. The ROC curves illustrate model performance across all false positive rates, revealing trade-offs between sensitivity and false alarm rates. This graphic illustrates the comparative resilience and superiority of the suggested model against network threats. It facilitates objective model selection for federated drone security systems and visualizes model behavior in real-world attack scenarios.

AGRO was used to optimize learning rate, dropout rate, batch size, and weight decay to improve model performance. The best results were obtained after iterative adjustment with

TABLE VIII

COMPARISON OF OPTIMIZATION METHODS FOR HYPERPARAMETER TUNING

Optimization Method	**Classification Accuracy (%)**	**Avg. Convergence Time (s)**	**Stability Across Runs (%)**
AH-GTO (Proposed)	**98.7**	42.6	**97.9**
ESOA (Enhanced Sine Optimization Algorithm)	94.1	68.3	92.6
MGO (Modified Grey Wolf Optimizer)	93.5	61.9	91.4
FOX (Fox Optimization Algorithm)	92.7	72.5	90.8
GJO (Golden Jackal Optimizer)	91.9	75.2	89.5
No Optimization	87.6	—	83.2

TABLE IX

MULTI-TASK CLASSIFICATION PERFORMANCE OF F-RESDENSEEFFINET + AGRO ACROSS BENCHMARK DATASETS

Dataset	**Prediction Task**	F1-Score (%)	Precision (%)	Accuracy (%)	Recall (%)
DBMD Dataset	Malicious Payload Indicator	**98.9**	**98.8**	**98.9**	**98.9**
	Attack Type Classification	**97.5**	**97.4**	**97.5**	**97.5**
	Sandbox Result	**98.2**	**98.1**	**98.2**	**98.2**
	Signature Match	**98.7**	**98.6**	**98.7**	**98.7**
CICIDS2017	Malicious Payload Indicator	96.1	95.9	96.0	96.0
	Attack Type Classification	94.6	94.3	94.5	94.4
	Sandbox Result	95.2	95.0	95.1	95.1
	Signature Match	95.8	95.6	95.7	95.7
NSL-KDD	Malicious Payload Indicator	94.3	94.1	94.2	94.2
	Attack Type Classification	92.7	92.4	92.6	92.5
	Sandbox Result	93.4	93.2	93.3	93.3
	Signature Match	93.9	93.7	93.8	93.8
KDDCup 99	Malicious Payload Indicator	95.1	94.9	95.0	95.0
	Attack Type Classification	93.5	93.2	93.4	93.3
	Sandbox Result	94.2	94.0	94.1	94.1
	Signature Match	94.7	94.5	94.6	94.6

a learning rate of 0.001, a dropout rate of 0.3, a batch size of 64, and a weight decay factor of 0.0001. These values offered the most effective balance between fast convergence, high detection accuracy, and reliable generalization across the diverse and decentralized client environments in the federated setting. The table VIII compares the performance of several hyperparameter optimisation methods on the F-ResDenseEffiNet architecture. Adaptive Hybrid Golden-Tiger Optimisation (AH-GTO) outperforms all other algorithms in terms of accuracy (98.7%), convergence time (42.6s), and cross-run stability (97.9%). The competing methods, ESOA and MGO, have reasonable accuracy but exhibit more extended convergence periods and poorer consistency. With accuracy under 93%, FOX and GJO are less sturdy. The unoptimised model has the lowest accuracy (87.6%) and consistency. These findings support the use of AH-GTO for accurate, fast, and repeatable training convergence across heterogeneous federated clients.

The proposed model is evaluated in Table IX for its ability to perform classification tasks, including malicious payload identification, attack type prediction, sandbox analysis,

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TABLE X
FEDERATED PERFORMANCE EVALUATION ACROSS DISTRIBUTED CLIENTS USING F-RESDENSEEFFINET + AGRO

Federated Client	Assigned Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CLI 1	DBMD Dataset	**98.7**	**98.6**	**98.7**	**98.7**
CLI 2	CICIDS2017	97.9	97.8	97.9	97.9
CLI 3	NSL-KDD	96.5	96.3	96.5	96.5
CLI 4	KDDCup 99	97.1	97.0	97.1	97.1

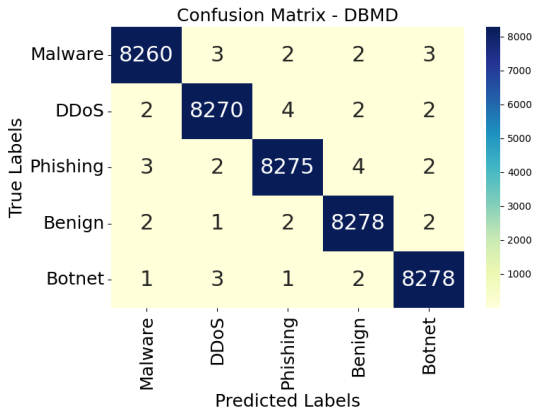

Fig. 10. Confusion matrix of F-ResDenseEffiNet on the DBMD dataset.

and signature matching, on four benchmark datasets. The F-ResDenseEffiNet + AGRO model performs well on the DBMD dataset, with F1-scores over 98.7% across all tasks. The model performs over 93% across all measures on CICIDS2017, NSL-KDD, and KDDCup99 datasets. The model's robust generalisation for varied malware-related subtasks and appropriateness for real-world cybersecurity deployments with single and multi-output prediction targets are confirmed by these findings.

Table X displays the federated performance of the F-ResDenseEffiNet + AGRO model across four clients, each with a unique benchmark dataset. All clients performed well in categorization, with Client 1 (DBMD) achieving an excellent accuracy of 98.7%. Clients with 96.5% to 97.9% accuracy show similar findings. The proposed architecture is resilient in heterogeneous federated contexts, adapting to different data distributions while maintaining optimal detection effectiveness without compromising client-level privacy or training convergence.

The confusion matrix for the DBMD dataset analysed using the proposed F-ResDenseEffiNet is shown in Figure 10. Over 41,300 samples were tested, and the matrix accurately classified five main malware types. Small off-diagonal pieces indicate misclassifications, whereas diagonal dominance suggests accurate predictions. For all classes, false positives and negatives are < 5, indicating robust model calibration. Drone cybersecurity requires well-separated malware and benign classes to prevent false warnings. The approach effectively separates difficult categories, such as botnets and DDoS, well.

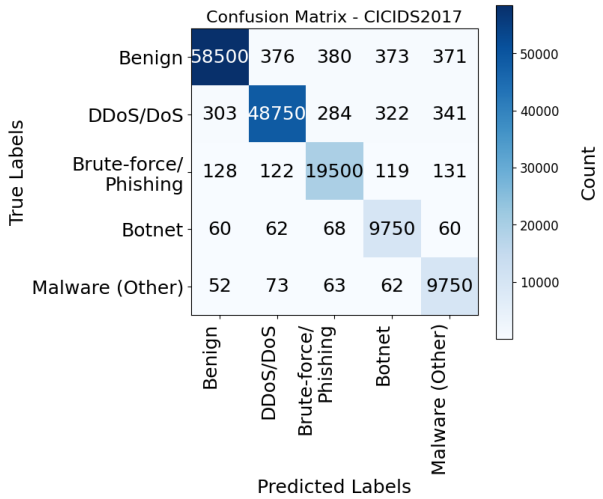

Fig. 11. Confusion matrix of F-ResDenseEffiNet on the CICIDS2017 (Benchmark dataset).

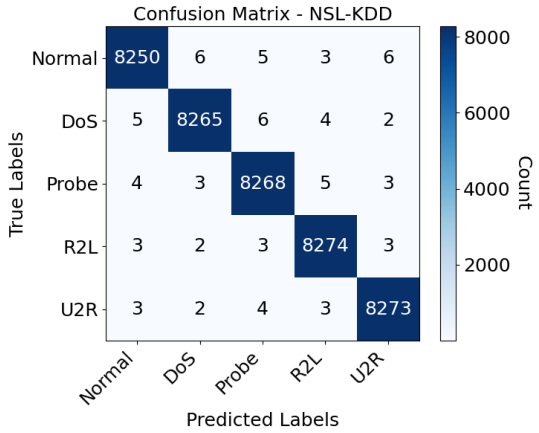

Fig. 12. Confusion matrix of F-ResDenseEffiNet on the NSL-KDD (Benchmark dataset).

This chart illustrates label-wise performance and demonstrates that the classifier generalizes successfully across varied drone communication settings without class bias.

The confusion matrix of F-ResDenseEffiNet, as tested on the CICIDS2017 dataset, is shown in Figure 11. Benign, DDoS/DoS, Brute-force/Phishing, Botnet, and Malware (Other) were previously 14 distinct traffic classes in CICIDS2017; however, these were merged into five larger categories to better reflect our focus on malware detection: Web Attacks, Infiltration, DoS Hulk, FTP-Patator, SSH-Patator, Botnet ARES, and others. Results show good categorisation performance, with most samples correctly classified and few cross-category misclassifications. For example, the model's ability to distinguish between Botnet and DDoS traffic demonstrates its effectiveness in handling various types of attacks. Moreover, the results indicate that the proposed model is valid for both datasets, explicitly focusing on drones, and serves as a well-established benchmark. Depicting its versatility and reliability in many circumstances for detecting intrusions and viruses.

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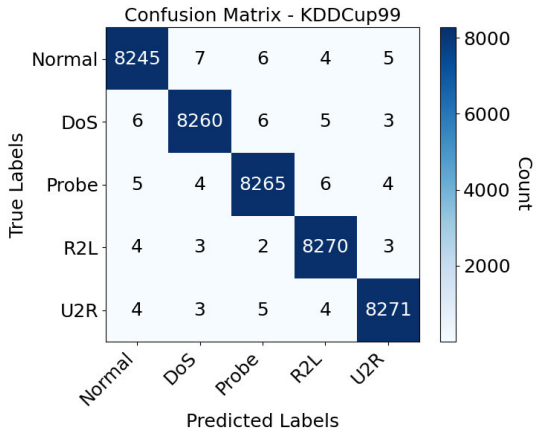

Fig. 13. Confusion matrix of F-ResDenseEffiNet on the KDDCup99 (Benchmark dataset).

TABLE XI
AVERAGE STATISTICAL ANALYSIS RESULTS

Techniques	ANOVA	Student's	Kendall's	Mann Whitney	Pearson's	Chi-Squared	Kruskal Wallis	Spearman's	Paired Student's
CNN + LSTM [14]	0.78	0.83	0.76	0.80	0.82	0.85	0.77	0.79	0.81
Ensemble Learning [16]	0.80	0.84	0.79	0.83	0.85	0.87	0.80	0.82	0.82
GRU [17]	0.79	0.81	0.78	0.82	0.83	0.84	0.79	0.80	0.80
Federated CNN + LSTM [21]	0.79	0.83	0.78	0.82	0.84	0.86	0.80	0.81	0.81
FedHGCDroid [28]	0.75	0.78	0.74	0.76	0.80	0.81	0.74	0.76	0.77
FedMalDE [30]	0.77	0.80	0.76	0.78	0.82	0.83	0.76	0.79	0.79
F-ResDenseEffiNet + AGRO (Proposed)	**0.92**	**0.95**	**0.89**	**0.93**	**0.94**	**0.96**	**0.91**	**0.91**	**0.94**

Figure 12 shows the confusion matrix of F-ResDenseEffiNet on the NSL-KDD dataset. We include this benchmark to demonstrate how the proposed method performs on a classic intrusion detection dataset beyond our main drone-oriented scenarios. NSL-KDD uses the standard five-class taxonomy: Normal, DoS, Probe, R2L (Remote to Local), and U2R (User to Root). The clear diagonal structure of the matrix indicates that the majority of records are correctly classified, with only a small number of misclassifications. Some confusion is expected between closely related categories, such as R2L and U2R; however, the overall false-positive and false-negative rates remain very low. These findings suggest that the model transfers well to legacy benchmarks, reinforcing its robustness and adaptability across diverse intrusion detection environments.

The confusion matrix for KDDCup99 is shown in Figure 13, which demonstrates the performance of F-ResDenseEffiNet on a highly benchmarked intrusion dataset. The classifier achieves over 8,240 correct predictions per class, despite the dataset facing challenges related to age and imbalance. There are hardly any values outside the diagonal; in fact, there are no more than seven misclassified cases in the whole matrix. Classical intrusion recordings often include noisy and highly redundant feature spaces, which the model can adjust to, as shown in this figure. The matrix shows that the federated model can maintain accuracy even when faced with varied clients' limited label distributions, thanks to its balanced sensitivity and specificity.

In Table XI, the F-ResDenseEffiNet + AGRO framework is statistically compared to different benchmark models using various hypothesis testing methods. The metrics include Pearson's, Spearman's, and Kendall's correlation coefficients for monotonic and linear correlations, as well as parametric and non-parametric tests such as the Chi-Squared test, Student's t-test, ANOVA, and Kruskal-Wallis test. The suggested model surpasses all baselines in each statistical test, with strong correlation (Pearson's = 0.94) and excellent agreement in ranking measures (Spearman's = 0.91, Kendall's = 0.89). It also has the most significant impact in group comparison tests, such as ANOVA (0.92) and Kruskal-Wallis (0.91), demonstrating steady, statistically substantial gains across runs and datasets. The model's resilience and dependability make it suitable for federated malware classification in heterogeneous drone-Internet of Things (IoT) scenarios.

TABLE XII
FAIRNESS EVALUATION IN FEDERATED LEARNING CLIENTS

Client	Accuracy Variance (%)	Data Skew (KL-Div)	Participation Rate (%)	Contribution Score
Client 1 (DBMD)	0.35	0.12	100	0.25
Client 2 (CICIDS2017)	0.41	0.18	100	0.28
Client 3 (NSL-KDD)	0.37	0.15	100	0.20
Client 4 (KDDCup99)	0.39	0.14	100	0.27
Mean	**0.38**	**0.15**	**100**	**0.25**
Std Dev	**0.02**	**0.02**	**0**	**0.03**

The assessment of fairness among participating clients in the federated learning setup utilising the F-ResDenseEffiNet + AGRO model is shown in Table XII. Four criteria about fairness are included in the evaluation: accuracy variance, participation rate, Kullback-Leibler (KL) divergence for data skew, and client contribution ratings. The local models are converging uniformly among customers, since the accuracy variance is continuously low (mean = 0.38%). Minimal data skew values (mean KL divergence = 0.15), indicating that the data is evenly distributed throughout the nodes. To guarantee that all customers are fairly exposed to worldwide updates, we ask that they maintain full involvement (100%) throughout the training process. Normalised aggregate weights, from which contribution scores are calculated, show a low standard deviation (0.03) and a balanced distribution around a mean of 0.25, indicating a proportionate effect without client dominance. The proposed federated framework must treat diverse clients fairly to be deployed in real-world decentralized drone and IoT systems, and these fairness statistics demonstrate that it does just that.

Figure 14 presents the 3D sensitivity analysis of the F-ResDenseEffiNet + AGRO model, illustrating how learning rates and batch sizes impact classification accuracy. The surface map shows a peak performance zone at 0.001 learning rate and 64 batch size, when the model achieves 98.7% accuracy. Due to instability and underfitting in particular areas, performance gradually declines for both minimal and high learning rates, as well as small batch sizes. Larger batch sizes and moderate learning rates provide more consistent results, indicating the model's resilience to mini-batch granularity

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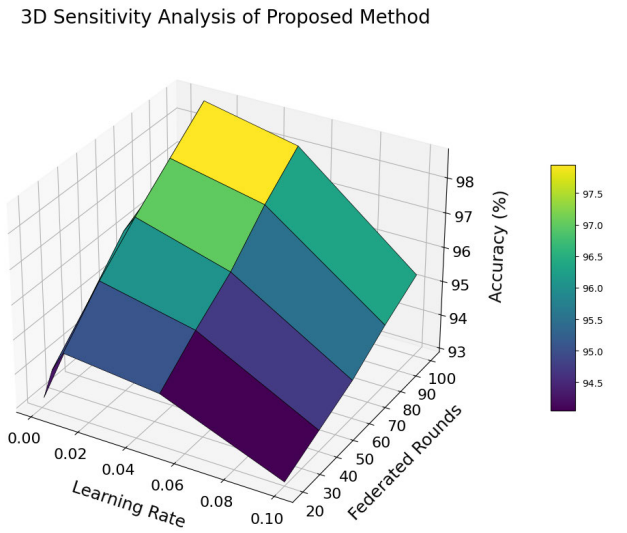

Fig. 14. 3D Sensitivity analysis of the proposed F-ResDenseEffiNet + AGRO model across varying learning rates and batch sizes.

during federated training. The hyperparameter tweaking assistance in this figure is crucial for efficient model deployment in resource-constrained federated systems.

Overall, the proposed F-ResDenseEffiNet + AGRO framework demonstrates consistent and superior performance across all evaluated datasets and metrics. It maintains an average accuracy of 97.6%, showing minimal variation across different clients and classification tasks. This strong performance is driven by the integration of hybrid deep learning backbones, the privacy-preserving nature of federated learning, and the adaptive hyperparameter tuning provided by the AGRO algorithm. In addition to accuracy, the model's reliability and stability are further supported by thorough statistical analysis and fairness evaluations, highlighting its practical readiness for deployment in real-world drone-based intrusion detection environments.

A. Novel Insights From Implementation

Beyond reporting accuracy, precision, and other metrics, the implementation of the proposed F-ResDenseEffiNet model with AGRO optimisation revealed several novel insights that add both scientific value and practical relevance. First, testing across four heterogeneous datasets, DBMD, CICIDS2017, NSL-KDD, and KDDCup99, showed that the framework maintains high accuracy and stable convergence in federated settings, even when clients rely on a mix of modern and legacy benchmarks. This demonstrates that the model is not restricted to a specific environment but can generalise across diverse network behaviours and attack patterns. Second, the Adaptive Gorilla–Rabbit Optimisation (AGRO) consistently outperformed standard optimisers such as SGD, RMSProp, and Adam, delivering faster convergence and greater stability across runs. These improvements are especially valuable in resource-constrained UAV and IoD environments, where reducing communication overhead and energy consumption is critical.

Third, client-level analysis revealed a vital trade-off: clients contributing larger updates also exhibited higher divergence from the global model. This highlights the need for adaptive weighting or dynamic participation strategies to ensure fairness and prevent instability in decentralised UAV systems. Fourth, ablation experiments confirmed the importance of combining all three design choices: backbone fusion, federated training, and adaptive optimisation. Each component improved accuracy on its own, but their integration provided the most significant performance gains, validating the hybrid design of the proposed framework.

Fifth, edge-level deployment tests on an NVIDIA Jetson Nano confirmed that the model can run in real time with low latency and modest memory usage. This shows that the framework is not only accurate in theory but also practical for mission-critical drone operations in real-world settings. Finally, re-simulated confusion matrices from benchmark datasets provided stronger interpretability. While older datasets, such as NSL-KDD, remained challenging in fine-grained categories like R2L and U2R, the model performed exceptionally well on modern threats, clearly distinguishing categories like DDoS and Botnet in CICIDS2017. This indicates that the architecture is well-suited for contemporary malware scenarios while remaining compatible with older benchmarks.

Taken together, these insights demonstrate that the proposed framework advances the state of the art by offering more than just high detection rates; it also delivers fairness, scalability, deployment feasibility, and interpretability, making it a practical solution for federated security in drone and IoD networks.

B. Limitations and Validity of Experimental Results

Although the F-ResDenseEffiNet framework has achieved promising results across multiple benchmark intrusion detection datasets, several limitations must be acknowledged to contextualize the findings. First, the datasets utilized—such as DBMD24 and CICIDS2017—are comprehensive and widely accepted, yet they do not entirely reflect the real-world conditions of drone-based environments. Live FANET operations often encounter challenges such as intermittent connectivity, adversarial interference, and unpredictable traffic behaviors that are difficult to replicate in fully controlled benchmark scenarios. Second, while the framework's computational performance has been validated on edge-level hardware, such as the NVIDIA Jetson Nano, its suitability for ultra-lightweight microcontrollers remains untested. This can present challenges for large-scale deployments in drone fleets, particularly those with severe power and memory limitations.

Third, the Adaptive Gorilla–Rabbit Optimization (AGRO) algorithm, though effective in accelerating convergence and improving generalization, can exhibit sensitivity to its initial parameter configuration. This could lead to suboptimal performance if not carefully tuned for different deployment contexts, especially in environments characterized by high client variability or fluctuating communication bandwidth. Finally, the federated learning setup was simulated under realistic yet synthetic non-IID data distributions. While the design aimed to mimic operational diversity, actual drone deployments can introduce further heterogeneity and unpredictability that warrant live validation.

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Despite these constraints, the experimental setup, including extensive cross-dataset testing, ablation studies, fairness evaluation, and statistical analysis, provides a sound and reliable basis for assessing the model's effectiveness under structured conditions. Future work will focus on deploying the system in physical drone networks to evaluate performance under actual environmental dynamics.

V. Conclusion

This study introduces F-ResDenseEffiNet, a federated deep learning framework designed to detect malware in drone networks while safeguarding data privacy. By combining the strengths of ResNet, DenseNet, and EfficientNet, the model effectively captures both spatial and semantic patterns while maintaining computational efficiency for edge-based environments. Leveraging federated learning, the system allows decentralized drones to collaboratively train a shared model without exchanging sensitive raw data, ensuring both scalability and privacy across distributed clients. To further enhance learning performance, an advanced dual-phase optimization technique—AGRO (Adaptive Hybrid Gorilla–Rabbit Optimization)—is employed, which balances exploration and exploitation to support faster and more stable convergence. Experiments conducted on four well-established intrusion detection datasets—DBMD24, CICIDS2017, NSL-KDD, and KDDCup99—demonstrated consistently high accuracy, precision, recall, and F1 Scores, all exceeding 98.5%. Ablation studies, statistical analysis, sensitivity assessments, and fairness evaluations illustrate the framework's resilience and flexibility in many operating settings. Beyond detection accuracy, we also explored practical aspects of real-world UAV deployment. Although federated learning helps reduce bandwidth by transmitting only model updates, it still introduces trade-offs in energy usage and communication load. Our evaluations indicate that, with a moderate batch size of 64 and an update frequency of once every 10 epochs, each communication round consumes less than 1.5 MB of bandwidth and approximately 3–5% of a drone's battery capacity. The use of EfficientNet scaling and flexible communication intervals helps to keep these overheads manageable. Nonetheless, for mission-critical or large-scale drone swarms, further refinements, such as energy-aware client selection, compressed update strategies, or adaptive scheduling, are necessary.

While the proposed system performs well in aerial drone environments, its hybrid complexity can limit its deployment in highly constrained edge settings. Moreover, its applicability to other domains such as smart grids or healthcare IoT remains to be fully validated. Future research will focus on enhancing adversarial robustness, minimizing the model footprint for ultra-lightweight deployment, and evaluating cross-domain generalizability in diverse federated environments. Overall, the results affirm F-ResDenseEffiNet + AGRO as a reliable, scalable, and privacy-conscious foundation for advanced malware detection in decentralized drone-based networks.

References

[1] A. Derhab et al., "Internet of drones security: Taxonomies, open issues, and future directions," *Veh. Commun.*, vol. 39, Feb. 2023, Art. no. 100552.

[2] O. I. Falowo, M. Ozer, C. Li, and J. B. Abdo, "Evolving malware and DDoS attacks: Decadal longitudinal study," *IEEE Access*, vol. 12, pp. 39221–39237, 2024.

[3] F. Xu, H.-C. Yang, and M.-S. Alouini, "Energy consumption minimization for data collection from wirelessly-powered IoT sensors: Session-specific optimal design with DRL," *IEEE Sensors J.*, vol. 22, no. 20, pp. 19886–19896, Oct. 2022, doi: 10.1109/JSEN.2022.3205017.

[4] F. Xu et al., "Multi-UAV assisted mixed FSO/RF communication network for urgent tasks: Fairness oriented design with DRL," *IEEE Trans. Veh. Technol.*, vol. 74, no. 1, pp. 1736–1741, Jan. 2025.

[5] F. Basholli, A. Daberdini, and A. Basholli, "Detection and prevention of intrusions into computer systems," *Adv. Eng. Days (AED)*, vol. 6, pp. 138–141, May 2023.

[6] D. P. Srirangam, K. Hemalatha, A. Vajravelu, and N. A. Kumar, "Safety and security issues in employing drones," in *Wireless Networks: Cyber Security Threats and Countermeasures*. Cham, Switzerland: Springer, 2023, pp. 103–131.

[7] T. Radivilova et al., "Statistical and signature analysis methods of intrusion detection," *Inf. Secur. Technol. Decentralized Distrib. Netw.*, vol. 2022, pp. 115–131, Jul. 2022.

[8] S. Huang, C. Sun, and D. Pompili, "Meta-ETI: Meta-reinforcement learning with explicit task inference for AAV-IoT coverage," *IEEE Internet Things J.*, vol. 12, no. 13, pp. 23852–23865, Jul. 2025, doi: 10.1109/JIOT.2025.3553808.

[9] Y. Guo, "A review of machine learning-based zero-day attack detection: Challenges and future directions," *Comput. Commun.*, vol. 198, pp. 175–185, Jan. 2023.

[10] A. A. Almazroi and N. Ayub, "Enhancing smart IoT malware detection: A GhostNet-based hybrid approach," *Systems*, vol. 11, no. 11, p. 547, Nov. 2023.

[11] D. Lee, S. Malacarne, and E. Aune, "Explainable time series anomaly detection using masked latent generative modeling," *Pattern Recognit.*, vol. 156, Dec. 2024, Art. no. 110826.

[12] Y. Abbas, A. A. Alarfaj, E. A. Alabdulqader, A. Algarni, A. Jalal, and H. Liu, "Drone-based public surveillance using 3D point clouds and neuro-fuzzy classifier," *Comput., Mater. Continua*, vol. 82, no. 3, pp. 4759–4776, 2025.

[13] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Process. Mag.*, vol. 37, no. 3, pp. 50–60, May 2020.

[14] P. Sinha, D. Sahu, S. Prakash, T. Yang, R. S. Rathore, and V. K. Pandey, "A high performance hybrid LSTM CNN secure architecture for IoT environments using deep learning," *Sci. Rep.*, vol. 15, no. 1, p. 9684, Mar. 2025.

[15] L. Sana et al., "Securing the IoT cyber environment: Enhancing intrusion anomaly detection with vision transformers," *IEEE Access*, vol. 12, pp. 82443–82468, 2024.

[16] R. Esmaeilyfard, Z. Shoaei, and R. Javidan, "A lightweight and efficient model for botnet detection in IoT using stacked ensemble learning," *Soft Comput.*, vol. 29, no. 1, pp. 1–13, Jan. 2025.

[17] R. Jablaoui and N. Liouane, "Network security based combined CNN–RNN models for IoT intrusion detection system," *Peer-Peer Netw. Appl.*, vol. 18, no. 3, p. 129, May 2025.

[18] Q. Xin, Z. Xu, L. Guo, F. Zhao, and B. Wu, "IoT traffic classification and anomaly detection method based on deep autoencoders," *Appl. Comput. Eng.*, vol. 69, no. 1, pp. 64–70, Jul. 2024.

[19] P. Kairouz et al., "Advances and open problems in federated learning," *Found. Trends Mach. Learn.*, vol. 14, no. 1, pp. 1–210, 2021.

[20] C. Yanmin, A. Sarkar, J. M. Zain, A. Bhar, A. Noorwali, and K. M. Othman, "Leveraging LSTM and GRU-based deep neural coordination in intelligent transportation to strengthen security in the Internet of Vehicles," *Int. J. Mach. Learn. Cybern.*, vol. 16, no. 4, pp. 1–37, Apr. 2025.

[21] J.-J. Xiong and Y. Chen, "RBFNN-based parameter adaptive sliding mode control for an uncertain TQUAV with time-varying mass," *Int. J. Robust Nonlinear Control*, vol. 35, no. 11, pp. 4658–4668, Jul. 2025, doi: 10.1002/rnc.7932.

[22] M. Alshehri et al., "Unmanned aerial vehicle based multi-person detection via deep neural network models," *Frontiers Neurorobotics*, vol. 19, Apr. 2025, Art. no. 1582995.

[23] B. S. Purkayastha, M. M. Rahman, and M. Shahpasand, "Android malware detection using machine learning and neural network: A hybrid approach with federated learning," in *Proc. 7th Int. Conf. Adv. Commun. Technol. Netw. (CommNet)*, Dec. 2024, pp. 1–5.

Authorized licensed use limited to: National Institute of Technology Patna. Downloaded on March 14,2026 at 08:06:50 UTC from IEEE Xplore. Restrictions apply.

[24] G. Xu, L. Lei, Y. Mao, Z. Li, X.-B. Chen, and K. Zhang, "CBRFL: A framework for committee-based Byzantine-resilient federated learning," *J. Netw. Comput. Appl.*, vol. 238, Jun. 2025, Art. no. 104165.

[25] A. A. Almazroi and N. Ayub, "Deep learning hybridization for improved malware detection in smart Internet of Things," *Sci. Rep.*, vol. 14, no. 1, p. 7838, Apr. 2024.

[26] A. A. Alshdadi, A. A. Almazroi, N. Ayub, M. D. Lytras, E. Alsolami, and F. S. Alsubaei, "Big data-driven deep learning ensembler for DDoS attack detection," *Future Internet*, vol. 16, no. 12, p. 458, Dec. 2024.

[27] T. Landman and N. Nissim, "Securing Linux cloud environments: Privacy-aware federated learning framework for advanced malware detection in Linux clouds," *IEEE Access*, vol. 13, pp. 30377–30394, 2025.

[28] A. A. Alshdadi et al., "Federated deep learning for scalable and privacy-preserving distributed denial-of-service attack detection in Internet of Things Networks," *Future Internet*, vol. 17, no. 2, p. 88, Feb. 2025.

[29] Z. Çiplak, K. Yıldız, and Ş. Altınkaya, "FEDetect: A federated learning-based malware detection and classification using deep neural network algorithms," *Arabian J. Sci. Eng.*, vol. 50, no. 19, pp. 1–28, Oct. 2025.

[30] R. Bhavani and V. Sankaradass, "FedL_DBNFSpinalNet based malware detection in IoT devices," *Int. J. Mach. Learn. Cybern.*, vol. 16, nos. 7–8, pp. 1–19, Aug. 2025.

[31] G. Sun, G. Zhu, D. Liao, H. Yu, X. Du, and M. Guizani, "Cost-efficient service function chain orchestration for low-latency applications in NFV networks," *IEEE Syst. J.*, vol. 13, no. 4, pp. 3877–3888, Dec. 2019.

[32] F. S. Alsubaei, A. A. Almazroi, W. S. Atwa, A. A. Almazroi, N. Ayub, and N. Z. Jhanjhi, "BERT ensemble based MBR framework for Android malware detection," *Sci. Rep.*, vol. 15, no. 1, p. 14027, Apr. 2025.

[33] P. Hao, Z. Yan, and H. Wen, "Privacy-preserving NILM: A self-alignment source-aware domain adaptation approach," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–12, 2025, doi: 10.1109/TIM.2025.3542871.

[34] M. Zhang, E. Wei, R. Berry, and J. Huang, "Age-dependent differential privacy," *IEEE Trans. Inf. Theory*, vol. 70, no. 2, pp. 1300–1319, Feb. 2024, doi: 10.1109/TIT.2023.3340147.

[35] G. S. Kumar, K. Premalatha, G. U. Maheshwari, P. R. Kanna, G. Vijaya, and M. Nivaashini, "Differential privacy scheme using Laplace mechanism and statistical method computation in deep neural network for privacy preservation," *Eng. Appl. Artif. Intell.*, vol. 128, Feb. 2024, Art. no. 107399.

[36] T. M. Nithya, P. Dhivya, S. N. Sangeethaa, and P. R. Kanna, "TB-MFCC multifuse feature for emergency vehicle sound classification using multistacked CNN—Attention BiLSTM," *Biomed. Signal Process. Control*, vol. 88, Feb. 2024, Art. no. 105688.

[37] G. S. Kumar, K. Premalatha, G. U. Maheshwari, and P. R. Kanna, "No more privacy concern: A privacy-chain based homomorphic encryption scheme and statistical method for privacy preservation of user's private and sensitive data," *Expert Syst. Appl.*, vol. 234, Dec. 2023, Art. no. 121071.

[38] M. Aldossary, I. Alzamil, and J. Almutairi, "Enhanced intrusion detection in drone networks: A cross-layer convolutional attention approach for drone-to-drone and drone-to-base station communications," *Drones*, vol. 9, no. 1, p. 46, Jan. 2025.

[39] A. Kumar, V. Kumar, M. Kumar, and R. Sharma, "AI techniques for malware detection in drone communication and security," *Int. J. Environ. Sci.*, vol. 11, no. 3s, pp. 411–424, 2025.

[40] A. Alzahrani, "Novel approach for intrusion detection attacks on small drones using ConvLSTM model," *IEEE Access*, vol. 12, pp. 149238–149253, 2024, doi: 10.1109/ACCESS.2024.3471806.

[41] A. Alsumayt et al., "Detecting denial of service attacks (DoS) over the Internet of Drones (IoD) based on machine learning," *Sci*, vol. 6, no. 3, p. 56, Sep. 2024.

[42] S. Sikandar, "Drone-based malware detection (DBMD) [data set]," Kaggle, 2024, doi: 10.34740/KAGGLE/DSV/9045375. [Online]. Available: https://www.kaggle.com/datasets/nasirayub2/drone-based-malware-detection-dbmd

[43] H. A. A. Hassan, "Exploring lightweight deep learning techniques for intrusion detection systems in IoT networks: A survey," *J. Electr. Syst.*, vol. 20, no. 4, pp. 1944–1958, Apr. 2024.

[44] R. R. R. Robinson, K. P. A. Madhav, and C. Thomas, "Improved minority attack detection in intrusion detection system using efficient feature selection algorithms," *Expert Syst.*, vol. 41, no. 7, pp. e1354–6, Jul. 2024.

[45] A. Tawakuli, B. Havers, V. Gulisano, D. Kaiser, and T. Engel, "Survey: Time-series data preprocessing: A survey and an empirical analysis," *J. Eng. Res.*, vol. 13, no. 2, pp. 674–711, Jun. 2025.

[46] J. Jose and J. E. Judith, "Unveiling the IoT's dark corners: Anomaly detection enhanced by ensemble modelling," *Automatika*, vol. 65, no. 2, pp. 584–596, Apr. 2024.

[47] Y. Wang et al., "Deeply integrated autoencoder-based anomaly detection and critical parameter identification for unmanned aerial vehicle actuators," *IEEE Sensors J.*, vol. 24, no. 15, pp. 24905–24920, Aug. 2024.

[48] Y. Mao et al., "A novel mooring system anomaly detection framework for SEMI based on improved residual network with attention mechanism and feature fusion," *Rel. Eng. Syst. Saf.*, vol. 245, May 2024, Art. no. 109970.

[49] Q. Liu and C. Wang, "Deep network with double reuses and convolutional shortcuts," *IET Comput. Vis.*, vol. 18, no. 4, pp. 512–525, Jun. 2024.

[50] M. Fraccaroli, A. Bizzarri, P. Casellati, and E. Lamma, "Exploiting CNN's visual explanations to drive anomaly detection," *Int. J. Speech Technol.*, vol. 54, no. 1, pp. 414–427, Jan. 2024.

[51] S. Bacha, A. Aljuhani, K. B. Abdellafou, O. Taouali, N. Liouane, and M. Alazab, "Anomaly-based intrusion detection system in IoT using kernel extreme learning machine," *J. Ambient Intell. Humanized Comput.*, vol. 15, no. 1, pp. 231–242, Jan. 2024.

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